

Before the Levy: Beef Prices During Denmark’s Cattle-Policy Transition

Diego González-González

29 September 2026

Abstract

How distinctive was Denmark’s beef-price movement during its 2024 cattle-policy transition? Comparing Danish beef-and-veal consumer prices with the same product in the other EU countries gives a post-June contrast of 0.016 log points (placebo standard error 0.029). With the full-sample synthetic weights held fixed, the contrast is 0.004 in the second half of 2024 and 0.024 in the first nine months of 2025. Beef instead rose by 0.064 log points relative to other Danish foods; that domestic contrast is concentrated in 2025 and its conventional significance depends on the serial-correlation bandwidth. Slaughter responses are imprecise. A model of costly adjustment of production plans gives a possible anticipatory mechanism and a conditional magnitude bound, but neither comparison isolates it. The country estimate is a post-June contrast conditional on an earlier path that already includes policy news; the concurrent nitrate-regulation change also prevents attribution to the future emissions tax alone. Grocery histories record price changes without a historical availability panel and cannot validate a monthly posted-price effect. The contribution is a same-product European benchmark and a timing account that substantially narrow, but do not identify, the Danish-specific price movement.

JEL codes: H23, Q18, Q54, Q58.

Keywords: Environmental taxation; policy announcements; anticipation; beef prices; livestock production; synthetic difference-in-differences.

1 Introduction

Environmental policy can change economic decisions well before it changes firms’ current tax liabilities. Producers expecting a future emissions charge may revise investment, procurement, and output plans when the policy is announced. Consumers may then face different prices during the years that separate agreement from implementation. Understanding this interval matters for the evaluation of climate policy: estimates beginning with the first tax payment can miss part of the adjustment that the policy has already induced.

I study beef-market outcomes during Denmark’s 2024 cattle-policy transition. The empirical question is how much of the apparent Danish price movement survives comparison with the same

product elsewhere, when that divergence appears, and what changes in slaughter accompany it. A competitive model in which changing production plans is costly supplies one possible interpretation: higher expected future charges can bring output adjustment forward. A fixed-stock slaughter-timing model gives the opposite price sign. The models organize the evidence rather than identify its cause.

Denmark’s Green Tripartite agreement provides the setting. On 24 June 2024, the government and negotiating parties agreed to introduce a tax on livestock greenhouse-gas emissions from 2030 (Danish Government, 2024). The expected liability was not necessarily entirely new information: an expert group had published agricultural carbon-tax models in February, and a parliamentary implementation agreement followed in November (Danish Ministry of Taxation, 2024a; Danish Ministry of Economic Affairs, 2024). Beef is the focus because cattle production involves substantial emissions and biological adjustment lags (Rosen et al., 1994). The April announcement and July end of the cattle nitrate derogation create a separate, overlapping cost shock (Danish Ministry of Environment, 2024). Accordingly, the empirical post indicator describes this policy episode; it cannot identify the isolated effect of the June emissions-tax news.

The main comparison uses Eurostat’s harmonized beef-and-veal index for Denmark and the other EU countries. Its synthetic-DiD contrast is 0.016 log points with a 0.029 standard error. Comparing the same commodity addresses common European beef shocks more directly. Statistics Denmark’s monthly indices give a complementary within-country contrast of 0.064 log points for beef relative to foods outside the direct livestock-tax base; that contrast is concentrated in 2025 and is sensitive to the serial-correlation adjustment. The country comparison remains exposed to Danish-specific cattle shocks and policy news before July. Grocery histories provide product descriptions and a provisional level scenario, but their entries are changes in quoted prices rather than verified daily observations. Historical product availability cannot be recovered from the frozen snapshot, so I do not interpret its event-price regression as corroborating evidence on monthly posted prices.

To examine the production margin, I use official monthly bovine slaughter statistics. Denmark is compared with 26 other EU countries, all measured in the same commodity and units. The main synthetic-DiD estimate for slaughter weight is -0.066 log units, with a wide interval. The estimate is close to zero with a longer, seasonally adjusted pre-period. Counts of slaughtered cattle provide similarly imprecise evidence. These results leave a substantial range of production responses compatible with the data and qualify the interpretation of the positive price estimates.

The temporal estimand is the July 2024–September 2025 price contrast relative to a weighted April 2023–June 2024 path, conditional on the comparison series. It is not the total effect of the cattle-policy transition: February tax proposals and the April announcement of the end of the EU Nitrates Directive derogation were already public before July. The nitrate change took effect on 31 July and could increase costs for some cattle farms (Danish Ministry of Environment, 2024). The November implementation agreement and European beef-market developments further complicate a reading of the 2025 divergence. Cross-country comparison removes neither Danish-specific shock. I therefore treat the estimates as descriptive, conditional contrasts.

The model adds a conditional quantitative restriction. Under stationary two-date primitives, the announcement-date market-price response cannot exceed a discounted fraction of the expected future output wedge. Mapping the two official retail-price coefficients through the same provisional grocery level produces different implications: the national food contrast exceeds the illustrative bound, while the same-product country contrast does not. Neither is a model test, because the retail and producer stages, taxable emissions, the level anchor, and the policy-attributable share are unknown. The separate social-cost-of-carbon accounting is retained in Appendix C as scenario arithmetic, not evidence on price formation or welfare.

The analysis relates to anticipation of environmental regulation (Smulders et al., 2012; Rittenhouse and Zaragoza-Watkins, 2018) and transmission of emissions costs into product prices (Fabra and Reguant, 2014). Its contribution is an explicit hierarchy of price counterfactuals, comparable timing summaries, and producer-side evidence for a closely spaced policy transition. Section 2 develops the model; Sections 3 and 4 describe measurement and comparisons; Section 5 presents the evidence and conditional magnitude exercise. Sections 6 and 7 discuss interpretation. Appendix C contains the environmental-price accounting.

2 Theoretical Framework

2.1 A credible future tax and planned supply

There are two decision dates: the announcement date, normalized to 0, and the implementation date $T > 0$. Let $Q_t > 0$ denote aggregate beef output at date $t \in \{0, T\}$, let $P_t(Q_t)$ be inverse demand with $P'_t(Q_t) < 0$, and let $C_t(Q_t)$ be the industry's private production cost with $C'_t(Q_t) > 0$ and $C''_t(Q_t) \geq 0$ for $Q_t > 0$. For an arbitrary reference quantity $\bar{Q}_t > 0$, define the competitive-market potential net of production cost as

$$G_t(Q) \equiv \int_{\bar{Q}_t}^Q P_t(x) dx - C_t(Q). \quad (1)$$

The reference quantity changes only an additive constant. Thus $G'_t(Q) = P_t(Q) - C'_t(Q)$ and $G''_t(Q) = P'_t(Q) - C''_t(Q) < 0$. One unit of output generates $\chi > 0$ tons of CO2-equivalent emissions. The statutory tax at T is τ_T DKK per ton of CO2e. Under risk neutrality, the announcement creates the expected output wedge

$$\omega \equiv \chi E_0[\tau_T], \quad E_0[\tau_T] = \pi \tau \quad (2)$$

in the special case in which the tax equals $\tau > 0$ with implementation probability $\pi \in [0, 1]$ and zero otherwise. Here $E_0[\cdot]$ is the expectation conditional on information at announcement. The implementation-date output plan is committed before implementation uncertainty is resolved and cannot be conditioned on the realization of τ_T . Expected tax payments are therefore $\chi E_0[\tau_T] Q_T$.

The policy agreement need not move beliefs from a zero expected liability. If ω^- is the expected wedge immediately before new information and ω^+ afterward, the information innovation is $\Delta\omega =$

$\omega^+ - \omega^-$. Proposition 1 gives the sign of a local response evaluated at the prevailing prior equilibrium when $\Delta\omega > 0$. The finite bound below compares a stationary zero-wedge equilibrium with a positive-wedge equilibrium; it cannot be applied mechanically to an unknown Danish belief revision. Neither the February proposals nor the June and November agreements reveal ω^- or ω^+ in these data.

Production planning is intertemporal. Changing planned output between dates costs $\kappa(Q_T - Q_0)^2/2$, where $\kappa \geq 0$ summarizes biological lags, capacity adjustment, and procurement commitments. Let $\beta \in (0, 1]$ be the one-period discount factor and $\delta \equiv \beta^T$ the discount factor from announcement to implementation. Assume an interior maximizer exists. Because the following potential is strictly concave, that maximizer is unique:

$$G_0(Q_0) + \delta \left[G_T(Q_T) - \omega Q_T - \frac{\kappa}{2}(Q_T - Q_0)^2 \right]. \quad (3)$$

The potential is a device for representing competitive equilibrium. Firms take prices as given, and the first-order conditions equate those prices to private marginal production and adjustment costs. This interpretation assumes that the aggregate cost and adjustment schedules represent the industry's feasible allocation across firms. Here Q_t denotes planned market output, and κ summarizes the connection between plans across dates. The model abstracts from detailed herd accounting and foreign supply. Applying its price prediction to Danish retail markets consequently also requires that trade and distribution margins do not fully offset the domestic response.

Proposition 1 (Anticipatory contraction) *Suppose the solution to (3) is interior and $\kappa > 0$. A larger expected implementation wedge reduces planned output both before and at implementation and raises the announcement-period market price:*

$$\frac{\partial Q_0}{\partial \omega} < 0, \quad \frac{\partial Q_T}{\partial \omega} < 0, \quad \frac{\partial P_0}{\partial \omega} > 0. \quad (4)$$

The policy-induced level change is zero when the policy is not credible ($\pi = 0$). The marginal pre-implementation response is zero when adjustment is costless ($\kappa = 0$). With fixed adjustment curvature and demand and cost curvatures bounded away from zero, it converges to zero as implementation becomes arbitrarily distant ($T \rightarrow \infty$) when $\beta < 1$.

The proof is in Appendix A. To derive demand and supply from primitives, suppose preferences and technology are stationary across the two dates. A representative household consumes beef $q_t > 0$ and a numeraire good z_t . Its quasi-linear utility is

$$u(z_t, q_t) = z_t + \begin{cases} A q_t^{1-1/\varepsilon} / (1 - 1/\varepsilon), & \varepsilon \neq 1, \\ A \log q_t, & \varepsilon = 1, \end{cases} \quad (5)$$

where $A > 0$ is the taste shifter and $\varepsilon > 0$ is the absolute price elasticity of beef demand. The numeraire price is one and income is sufficient for an interior solution. Utility maximization subject

to $z_t + p_t q_t \leq y_t$, where p_t is the beef price and y_t is income, followed by market clearing $q_t = Q_t$, gives inverse demand

$$P(Q) = A Q^{-1/\varepsilon}. \quad (6)$$

Quasi-linearity isolates the commodity-market channel from income effects; it is not required for Proposition 1. The isoelastic schedule is interpreted locally around $Q^* > 0$, rather than as a literal description of willingness to pay near zero consumption.

On the production side, let a representative competitive producer use fixed capital $K > 0$ and a variable composite input $X_t \geq 0$ to produce

$$Q_t = B K^{1-\alpha} X_t^\alpha, \quad B > 0, \quad \alpha \in (0, 1), \quad (7)$$

where B is total factor productivity and α is the output elasticity of the variable input. If that input costs $v > 0$ per unit, conditional cost minimization gives

$$C(Q) = v \left(\frac{Q}{B K^{1-\alpha}} \right)^{1/\alpha}, \quad C'(Q) = m Q^\eta, \quad (8)$$

where $\eta \equiv (1 - \alpha)/\alpha > 0$ is the inverse supply elasticity and $m \equiv (v/\alpha)(B K^{1-\alpha})^{-1/\alpha} > 0$. Here K is quasi-fixed sectoral productive capacity: entry and major capacity expansion are suppressed over the model horizon. The function $C(Q_t)$ captures within-date variable cost conditional on that capacity, whereas $\kappa(Q_T - Q_0)^2/2$ captures additional dynamic reconfiguration costs not contained in X_t . Fixed capacity makes variable cost strictly convex. The no-tax, no-adjustment static equilibrium therefore has

$$Q^* = \left(\frac{A}{m} \right)^{1/(\eta+1/\varepsilon)}, \quad P^* = A(Q^*)^{-1/\varepsilon} = m(Q^*)^\eta. \quad (9)$$

Corollary 1 (Microfounded local incidence) *Under (5)–(8) and $\kappa > 0$, define $s \equiv \eta + 1/\varepsilon > 0$, the normalized adjustment curvature $\rho \equiv \kappa Q^*/P^* > 0$, and the normalized expected wedge $\hat{\omega} \equiv \omega/P^*$. At the stationary zero-wedge equilibrium $Q_0 = Q_T = Q^*$, the exact local comparative statics are*

$$\left. \frac{\partial \log Q_0}{\partial \hat{\omega}} \right|_{\hat{\omega}=0} = - \frac{\delta \rho}{s[s + \rho(1 + \delta)]}, \quad (10)$$

$$\left. \frac{\partial \log Q_T}{\partial \hat{\omega}} \right|_{\hat{\omega}=0} = - \frac{s + \delta \rho}{s[s + \rho(1 + \delta)]}, \quad (11)$$

$$\left. \frac{\partial \log P_0}{\partial \hat{\omega}} \right|_{\hat{\omega}=0} = \frac{\delta \rho}{\varepsilon s[s + \rho(1 + \delta)]} > 0. \quad (12)$$

Appendix A derives the corollary. Equation (12) is exact for a marginal wedge around the untaxed stationary equilibrium; finite changes obtained from it are first-order approximations, not global effects. Since $\omega = \pi \tau \chi$, the exact marginal response to the announced rate at $\tau = 0$ equals the

coefficient in (12) times $\pi\chi/P^*$. Its local marginal magnitude increases with credibility π , emissions intensity χ , proximity to implementation δ , and adjustment curvature ρ . Technology curvature and demand curvature determine the quantity response through s ; $1/\varepsilon$ additionally maps the output response into the price response. The result does not imply that the observed retail-price change equals the future tax wedge.

Corollary 2 (Two-date price bound) *Under stationary demand and costs, weakly increasing marginal production cost, an interior solution, and $\omega > 0$, the finite announcement-date response satisfies*

$$P_0 - P^* \leq \frac{\delta}{1 + \delta} \omega. \quad (13)$$

The bound concerns the model's market price and expected output wedge; it is not a bound on an observed retail-price change in an open, multi-period economy.

The proof is in Appendix A. The bound gives the model a quantitative implication: for $\delta \leq 1$, the announcement price response cannot exceed one half of the anticipated future output wedge under these assumptions.

2.2 From planned output to measured beef markets

The two-date Q_t suppresses several livestock margins. A stock-flow representation makes the distinction explicit. If H_t is the cattle stock, B_t births or additions, S_t slaughtered animals, μ other attrition, and w_t mean carcass weight, then

$$H_{t+1} = (1 - \mu)H_t + B_t - S_t, \quad Q_t = w_t S_t. \quad (14)$$

Domestic beef consumption can differ from Q_t through imports, exports, and inventory changes. Its retail price can differ from the domestic producer price through processing and distribution margins. Equation (14) shows why an initial herd-retention response could lower slaughter counts before changing carcass weight, and why an import response could attenuate the retail price effect. Dairy co-production makes births and retention depend on dairy-market conditions as well. The available data observe slaughter weight and heads but do not identify all of these flows or margins.

For an animal with taxable emissions e_i and deduction d_i , a simplified liability is $\tau \max\{e_i - d_i, 0\}$. Where $e_i > d_i$, the marginal incentive to reduce taxable emissions is τ , whereas the average liability per unit of output is $\tau(e_i - d_i)/q_i$ for output q_i . The two-date ω is closer to the latter on a fixed-intensity output margin. Gross lifecycle emissions can enter damages without equaling net taxable farm emissions. These distinctions prevent the model's single χ from being used simultaneously as a measured damage intensity and a measured tax base.

2.3 Environmental benchmark and a sign-reversing boundary case

Let environmental damages at date t be $D_t(E_t)$, where emissions are $E_t = \chi Q_t$, $D'_t(E_t) > 0$, and D_t is measured in DKK. Define $MD_t^e \equiv D'_t(E_t)$ as marginal damage per ton of CO2e. The standard

Pigouvian benchmark is

$$\tau_t^P = MD_t^e. \quad (15)$$

For the paper’s accounting exercise, I set $MD_t^e = SCC_t$ under an explicit one-for-one valuation of a ton of CO₂e at the CO₂ SCC. This convention is not a mechanical equivalence for methane and nitrous oxide merely because global-warming potentials express them in CO₂e.

Proposition 2 (Local welfare effect) *Suppose tax revenue is rebated lump-sum, administration is costless, private and social discount factors coincide, emissions intensity is fixed, adjustment costs are real resource costs, and the assumptions of Proposition 1 hold. Starting from no emissions price, a small increase in a credible future tax raises welfare whenever marginal environmental damage is positive. More generally, an increase in ω raises welfare locally if the expected output wedge does not exceed marginal implementation-date damage per unit of output, $\omega \leq \chi D'_T(E_T)$; the induced pre-implementation contraction supplies an additional benefit when $D'_0(E_0) > 0$.*

The welfare result concerns a marginal policy change from an initially unpriced externality under the model’s assumptions. Assessing the Danish rate requires a further distinction between output and emissions-intensity choices. The announced 2030 schedule has a DKK 300/tCO₂e marginal abatement incentive, while the per-animal deduction produces a DKK 120/tCO₂e average burden at the reference emissions intensity. Both amounts are stated in 2022 prices (Danish Government, 2024; Danish Ministry of Taxation, 2024b). For a representative animal with fixed reference intensity, the latter provides the closer approximation to the model’s output wedge. Farms with different emissions per animal face different average burdens.

Finally, the sign in Proposition 1 is not mechanical. Suppose instead that a fixed stock $\mathcal{S} > 0$ of slaughterable output, measured in the same units as Q_t , must be allocated across the two dates, so $Q_0 + Q_T = \mathcal{S}$, with an interior allocation $0 < Q_0 < \mathcal{S}$ and no margin for planned herd contraction. If the tax liability applies to later slaughtered output, a future tax makes later slaughter relatively less attractive, shifts slaughter to date 0, and therefore gives $\partial Q_0 / \partial \omega > 0$ and $\partial P_0 / \partial \omega < 0$. This pure reallocation result need not apply if emissions accrue continuously before slaughter. The preemptive-slaughter channel is the livestock analogue of an anticipation or green-paradox response (Smulders et al., 2012). A positive price response is compatible with planned contraction. The slaughter estimates below examine its quantity implication, while allowing for the distinct timing mechanism in the fixed-stock model.

3 Data and Panel Construction

3.1 Scraped grocery-price microdata

The grocery source is the historical catalog assembled by dagligepriser.dk. Its collector checks prices daily, but the retained history adds an entry when the price, quantity, or unit changes; unchanged daily quotes do not create entries. The available June 2026 snapshot also lacks a

historical availability record for every store and product. I therefore cannot distinguish a continuing unchanged price from an intervening exit or failed collection. Averaging entries within a product–store–month measures the average of recorded change events, not the time-weighted price paid during that month. A month with no change can disappear even if the product was continuously sold. The resulting regression is retained as a diagnostic of the event records and is not used as independent evidence on monthly posted prices. A validated price-spell panel would require dated availability or archived daily snapshots.

Prices are converted to DKK per kilogram for mass units and DKK per litre for volume units. Reported package quantities take precedence. When quantity or unit fields are absent, the procedure searches the product description for a number followed by kg, g, litre, ml, or cl. Grams, millilitres, and centilitres are converted to their base units before dividing price by quantity. Items measured only per piece and observations with missing, non-positive, or unsupported quantities are excluded. Because normalized prices remain right-skewed, estimation uses their natural logarithm and Table 1 reports medians and quartiles as well as means.

Candidate beef names are initially found with Danish and English meat terms, then screened against the public product descriptions using a six-category Jev classification and a conservative core-beef probability threshold. I inspected the retained names and excluded evident false positives such as beef-flavored noodles and sliced prepared beef. This is a reproducible screening rule, not an externally validated measure of classifier precision. Prepared dishes, plant substitutes, pet food, mixed livestock products, and uncertain candidate names are excluded. Livestock-exposed food products, including poultry, eggs, and dairy, are excluded from the control group. Package parsing now recognizes multipacks such as two 400 g units; reported mass or volume remains the first source for normalization. A classification audit cannot repair the absence of historical price spells or availability.

The event month, June 2024, is excluded from the grocery pre- and post-periods. The window is October 2023–September 2025, excluding June, and calendar event time retains an empty slot at zero; May is -1 and July is $+1$. The end date precedes the officially dated October 2025 bovine viral diarrhoea outbreak, although surveillance detected an affected dairy herd in September (Danish Veterinary and Food Administration, 2026; Dansk Veterinærtidsskrift, 2025). That endpoint does not rule out earlier market responses to the disease.

Among 151 screened beef product–store pairs with a recorded event on both sides of the announcement, events appear in 1,105 of 2,552 months between each pair’s first and last record, excluding June 2024 (43.3 percent). The corresponding share among 13,754 control pairs is 53.3 percent. These gaps cannot be labeled stockouts: unchanged posted prices and collector failures are also possible. They demonstrate why forward-filling the history would add an unverified availability assumption. The screen uses a saved review of 392 distinct names initially flagged as beef candidates; it retains 118 names after applying a 0.90 core-beef probability threshold and two explicit exclusions for prepared or ambiguous products. This is a reproducible name screen, not a validation study of the underlying price histories.

3.2 Official aggregate consumer prices

The second source is Statistics Denmark’s monthly consumer price index by COICOP 2018 product class. Beef and veal (011221) is the treated series. Candidate donors are terminal food and non-alcoholic-beverage classes with complete coverage in the estimation window. I exclude pork; poultry; eggs; lamb, sheep, and goat; dried or salted meat; offal; meat preparations; milk, yoghurt, cheese, cream, dairy desserts, other dairy products; and butter and other dairy fats. These categories are either directly livestock-exposed or sufficiently close to treated animal products that they could absorb the same announcement response. The resulting balanced panel contains beef and 50 donors.

The official-data window runs from April 2023 through September 2025: 15 months before July 2024 and 15 months from July 2024 onward. June 2024 is the final pre-treatment reference month for event-study purposes. This symmetric window prevents the aggregate estimates from being driven by unequal pre- and post-support and, like the microdata restriction, ends before the official October 2025 BVD outbreak interval.

3.3 Cattle slaughter and production

Eurostat provides monthly bovine slaughter statistics for all 27 EU Member States (Eurostat, 2026b). I measure output by carcass weight in thousand tonnes and by the number of animals slaughtered in thousand heads. The main panel covers April 2023–September 2025, matching the official consumer-price window. Each outcome has 810 country–month observations, with Denmark as the exposed country and 26 other Member States as potential comparisons. All countries have positive, non-missing observations throughout this window.

Slaughterhouse activity measures a production margin that can respond to both herd adjustment and the timing of slaughter. It differs from domestic cattle production when live animals cross borders. A separate monthly series from Statistics Denmark measures national cattle production, including the slaughter-equivalent contribution of live-animal trade (Statistics Denmark, 2026). I retain this distinction when interpreting the Eurostat estimates. Longer pre-announcement histories beginning in January 2020 permit a sensitivity analysis accounting for country-specific seasonality.

3.4 EU beef-price and trade comparisons

To address common beef-market movements, I use Eurostat’s monthly Harmonised Index of Consumer Prices for beef and veal (Eurostat, 2026a). All 27 EU Member States have positive, non-missing observations throughout April 2023–September 2025, giving 810 country–month observations. Denmark is exposed and the other 26 Member States form the donor pool. The Danish HICP is compiled by Statistics Denmark, so this changes the comparison group while retaining closely related underlying Danish consumer-price measurement. It is central to assessing the national food comparison, although cross-country spillovers and Denmark-specific cattle policies remain possible.

A separate check uses the European Commission’s Beef Trade Data Explorer, drawing on Eurostat COMEXT data under statistical regime 4. The data report extra-EU beef trade by Member State, external partner, month, and CN8 product. I aggregate imports to monthly carcass-weight tonnes and code unreported importer–partner–months as zero, an assumption that requires verification against reporting rules. The resulting panel contains 383 importer–partner pairs, including 17 involving Danish imports. Its average log-one-plus bilateral outcome is neither Denmark’s proportional change in total beef imports nor evidence about intra-EU trade.

Table 1 describes the price and slaughter samples. The grocery diagnostic contains 134,121 observed product–store–months from 13,905 pairs. These are months with recorded price-change events; historical product availability is unobserved. Normalized event-price levels are strongly right-skewed, which motivates logarithmic estimation and reporting medians alongside means. The official sample contains 30 monthly observations for each of 51 series. Its tighter distribution reflects the aggregation of price quotes within consistent product classes. Panel C describes monthly slaughter weight and counts in the production estimation sample. Differences in country size make the dispersion across countries informative about scale, while estimation uses logarithms and country-specific comparisons.

Table 1: Descriptive statistics for the estimation samples

Sample	Variable	Observations	Mean	SD	P25	Median	P75
<i>Panel A. Scraped product–store change-event prices</i>							
All	Normalized price	134,121	556.44	4,581.52	50.00	100.00	183.33
Beef	Normalized price	1,105	155.36	81.54	99.90	125.90	185.64
Controls	Normalized price	133,016	559.77	4,600.36	49.97	100.00	183.33
All	Months per product–store	13,905	9.65	6.12	4	8	15
Beef	Months per product–store	151	7.32	4.95	4	5	10
Controls	Months per product–store	13,754	9.67	6.13	4	8	15
<i>Panel B. Statistics Denmark consumer price indices</i>							
All	Consumer price index	1,530	97.82	6.87	95.25	98.76	101.24
Beef	Consumer price index	30	89.43	7.41	84.74	86.58	89.73
Donors	Consumer price index	1,500	97.99	6.75	95.51	98.82	101.26
<i>Panel C. Eurostat monthly bovine slaughter</i>							
All	Weight (1,000 tonnes)	810	19.86	28.10	1.34	5.72	32.78
Denmark	Weight (1,000 tonnes)	30	9.61	0.80	8.91	9.70	10.19
Donors	Weight (1,000 tonnes)	780	20.26	28.57	1.28	4.75	35.17
All	Animals (1,000 heads)	810	67.34	91.41	5.80	18.36	134.28
Denmark	Animals (1,000 heads)	30	35.98	2.99	33.56	36.24	38.14
Donors	Animals (1,000 heads)	780	68.55	92.94	5.57	16.33	147.45

Source: dagligepriser.dk (Panel A); Statistics Denmark (Panel B); Eurostat (Panel C).

Notes: Scraped prices are normalized to DKK per kilogram or litre and describe observed change-event months, not continuously available posted prices. For the months-per-product–store rows, observations count product–store units. Panel B covers the balanced April 2023–September 2025 window; its observation count is series–months. Panel C uses the same window and counts country–months. Price donors exclude livestock-exposed food categories; slaughter donors are the other EU countries.

4 Empirical Strategy

The supplementary grocery regression is

$$\log \tilde{p}_{it} = \alpha_i + \lambda_t + \beta^{events}(Beef_i \times Post_t) + \varepsilon_{it}, \quad (16)$$

where $\tilde{p}_{it}$ averages normalized change-event quotes for product–store pair i in month t . Product–store and month effects are α_i and λ_t . The coefficient β^{events} is a descriptive contrast among observed event months. Product–store clustered errors account for within-unit dependence, but there is one national policy episode and the missing-price process may vary with repricing. These errors are not uncertainty for a causal national price effect. The event plot omits May 2024; June remains an empty calendar-month slot.

For the official-data DiD, I subtract the unweighted mean log CPI of the 50 food donors from log beef CPI in each month and regress this gap on a post-period indicator. This yields 30 monthly regression observations from 1,530 underlying series–months. Two-lag Newey–West errors are the baseline short-dependence adjustment; I also report lags three, four, and six because a monthly beef-price gap may remain correlated longer. The monthly plot displays the beef-minus-food gap relative to June 2024 without pointwise intervals. With only one treated CPI series and a saturated treated-month block, robust residual-based intervals would not capture unrestricted treated-series shocks.

Synthetic DiD follows Arkhangelsky et al. (2021). To use common notation across the balanced panels, let y_{it} denote log CPI or HICP, log slaughter output, or log-one-plus imports, as appropriate, and let D_i identify exposed units. With $W_{it} = D_i Post_t$, the estimator solves

$$(\hat{\beta}^{SD}, \hat{\alpha}, \hat{\lambda}) \in \arg \min_{\beta, \alpha, \lambda} \sum_i \sum_t \hat{w}_i \hat{v}_t (y_{it} - \alpha_i - \lambda_t - \beta W_{it})^2. \quad (17)$$

Here α_i and λ_t retain their unit- and month-effect interpretation. Nonnegative donor weights $\hat{w}_i$ align pre-announcement outcome paths with the treated mean, allowing an intercept and regularizing the fit. Nonnegative pre-period weights $\hat{v}_t$ balance donors’ pre- and post-announcement outcomes, also allowing an intercept. Donor weights sum to one, as do pre-period weights; treated units and post-periods receive uniform weights within their respective groups.

Let $\mathcal{C}$ and $\mathcal{T}$ denote donor and treated units, and $\mathcal{P}$ and $\mathcal{A}$ the pre- and post-announcement periods. Write $N_1 = |\mathcal{T}|$, $T_1 = |\mathcal{A}|$, and $\bar{y}_{\mathcal{T}t} = N_1^{-1} \sum_{i \in \mathcal{T}} y_{it}$. The equivalent weighted-gap expression is

$$\hat{\beta}^{SD} = \frac{1}{T_1} \sum_{t \in \mathcal{A}} \left(\bar{y}_{\mathcal{T}t} - \sum_{i \in \mathcal{C}} \hat{w}_i y_{it} \right) - \sum_{t \in \mathcal{P}} \hat{v}_t \left(\bar{y}_{\mathcal{T}t} - \sum_{i \in \mathcal{C}} \hat{w}_i y_{it} \right). \quad (18)$$

Thus the post-announcement gap is adjusted for its weighted pre-announcement level. Uniform donor and pre-period weights recover the balanced-panel DiD comparison. The official CPI DiD remains the preferred within-Denmark price specification; synthetic weighting probes its dependence on donor paths. The country HICP comparison uses the same beef product across countries

and is economically preferable for controlling common European beef shocks. I report both because their identifying restrictions differ. Placebo inference for synthetic DiD requires comparable untreated errors across donor and treated units; repeated draws do not create new policy episodes.

Interpreting these coefficients as announcement effects requires that beef and the retained comparison foods would have followed parallel trends without the announcement, and that the controls did not respond to it. Excluding livestock products reduces direct exposure to the emissions charge. It does not remove substitution between foods or exposure to the agreement’s other agricultural measures. Product–store clustering also leaves a limitation: a national beef-specific shock can correlate errors across many stores and products. A large number of product–store observations does not provide an equally large number of independent policy assignments. The aggregate specifications offer a complementary view of uncertainty at the commodity level.

Synthetic DiD compares Denmark’s log beef-and-veal HICP with the same series in other Member States over the same pre- and post-periods. It uses placebo inference across countries. The trade diagnostic compares log-one-plus extra-EU imports across bilateral importer–partner pairs. Its unit-cluster bootstrap does not generate independent Danish policy shocks: all Danish pairs share the same exposure. The trade outcome is also unit-dependent because of the added constant and omits intra-EU imports. I therefore do not use its null result to rule out foreign substitution.

The production analysis applies synthetic DiD to log monthly bovine slaughter weight, with Denmark treated and 26 other EU countries as donors. Slaughter counts provide an alternative outcome. Appendix B.1 reports the sample, seasonal adjustment sensitivity, and June-omission check. This design avoids treating another Danish livestock category as unaffected. As in the country-price comparison, a Danish cattle-specific shock remains a potential confound.

In particular, the end of the cattle nitrate derogation was announced on 10 April 2024 and took effect on 31 July 2024 (Danish Ministry of Environment, 2024). It reduced the permitted nitrogen application from livestock manure for affected farms from 230 to 170 kg per hectare. The ministry explicitly anticipated higher costs for some cattle farms. This regulatory change can influence both prices and production during the estimation window. The available data do not identify its contribution separately from that of the livestock-tax announcement.

5 Results

Table 2 reports the grocery diagnostic first, as a transparent description of the original microdata design, followed by three official-data comparisons. The principal same-product country contrast is 0.016 log points (placebo standard error 0.029). Its 95 percent interval, $[-0.042, 0.073]$, includes zero and the within-Denmark coefficient; it neither establishes a Danish-specific increase nor demonstrates that the national movement was entirely common to Europe. The complementary beef-versus-food CPI DiD is 0.064 (HAC standard error 0.030), and synthetic weighting of those Danish series gives 0.076 (placebo standard error 0.046). These are different counterfactuals, not a formal test of their difference. Teagasc reported a 3.3 percent decline in EU beef supply in the

first quarter of 2025 and rising retail beef prices elsewhere in Europe, a plausible competing source of the later domestic contrast (Teagasc, 2025).

The country synthetic donor weights have a Herfindahl sum of squares of 0.124, equivalent to 8.1 equally weighted donors among 26; Finland, the Netherlands, and Czechia together receive 54.1 percent. Only four pre-period months have positive time weights. May and June 2024 jointly receive 78.5 percent, including 64.8 percent on June; the inverse sum of squared time weights is 2.16, a concentration diagnostic rather than an inferential sample size. Tax proposals and the nitrate announcement preceded those months, and the tax agreement arrived on 24 June. The estimated country contrast is therefore a post-June comparison conditional on part of the developing episode, not its total effect. Omitting June gives 0.022 (placebo standard error 0.028); the corresponding domestic CPI DiD is 0.064 (HAC standard error 0.030). Dropping each country donor in turn gives point contrasts from 0.006 to 0.026. Short pre-period holdouts give -0.009 for October–December 2023 and 0.004 for April–June 2024; the latter already contains policy news. These checks have no new standard errors and do not validate the post-period counterfactual.

Table 2: Beef-price contrasts around Denmark’s 2024 cattle-policy episode

	(1) Grocery change event DiD	(2) CPI DiD	(3) CPI SDiD	(4) EU beef HICP SDiD
ATT (descriptive contrast)	0.102*** (0.013)	0.064** (0.030)	0.076* (0.046)	0.016 (0.029)
Regression observations	134,121	30	1,530	810
Underlying event/series–months	134,121	1,530	1,530	810
Units	13,905	51	51	27
Periods	23	30	30	30
Pre-treatment beef level/index	147.77	84.88	84.88	122.21
HAC maximum lag	—	2	—	—
Time window	2023m10–2025m9	2023m4–2025m9	2023m4–2025m9	2023m4–2025m9

Source: dagligepriiser.dk (column 1); Statistics Denmark (columns 2–3); Eurostat HICP, compiled by national statistical institutes (column 4).

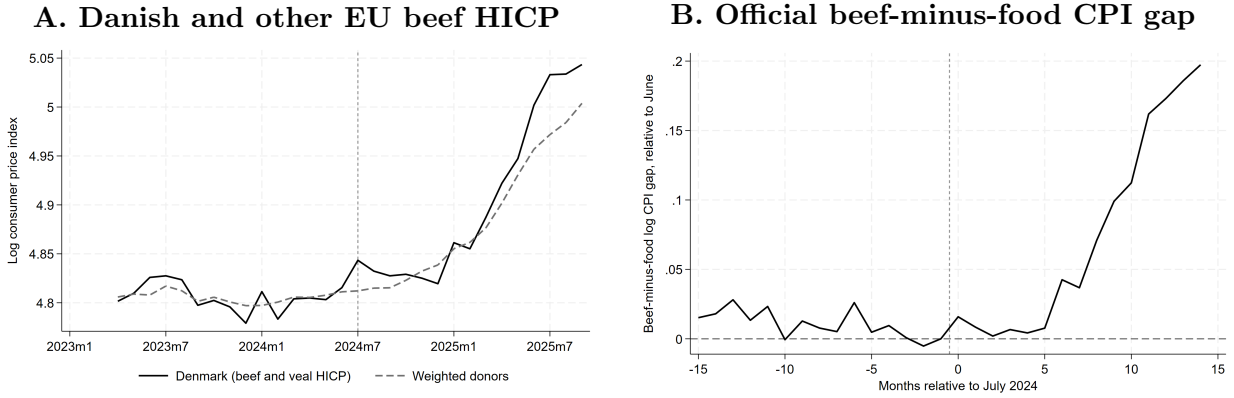
Notes: Column 1 measures log prices averaged across recorded change events in product–store–months; its 151 beef units and 1,105 beef event-months have within-window support on both sides of June 2024. Its first-column placement makes the diagnostic visible, not primary evidence on monthly posted prices. Product–store clustering describes variation conditional on event selection and does not supply national-policy inference. Columns 2–4 measure log consumer-price indices. Column 2 regresses 30 monthly beef-minus-food gaps on a post indicator and reports Newey–West HAC errors with two lags; columns 3–4 use synthetic DiD with 200 placebo replications. Columns 2–3 compare beef with 50 other Danish food classes; column 4 compares Denmark with 26 EU countries for beef and veal. Pre-treatment beef level is DKK/kg in column 1 and an index in columns 2–4. June 2024 is omitted only in column 1. Parentheses contain standard errors. Stars describe reference procedures, not causal identification: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The domestic CPI SDiD 95 percent interval is $[-0.014, 0.166]$. The CPI DiD result is sensitive to HAC lag truncation: keeping the 0.064 coefficient fixed, lags two, three, four, and six give standard errors 0.030, 0.034, 0.036, and 0.038, respectively, with two-sided p -values 0.044, 0.066, 0.082, and 0.098. Two lags are a transparent short-dependence baseline, not an empirically established cut-

off. With 30 monthly observations, these reference distributions are approximate; the substantive conclusion does not turn on crossing 0.05.

Figure 1 places the same-product European comparison beside the domestic beef-minus-food gap. Holding the full-sample country synthetic and pre-period time weights fixed, its post-June gap relative to the weighted pre-period is 0.004 in July–December 2024 and 0.024 in January–September 2025. These are descriptive segments of one fitted contrast, without separate standard errors. Relative to its full pre-period mean, the domestic log CPI gap averaged -0.003 and 0.109 in the same two segments. The domestic movement therefore arose in 2025, after the November implementation agreement and amid wider European beef-price pressure; the country divergence is smaller in both segments. The sequence is compatible with delayed adjustment but does not identify it.

Figure 1: Two official beef-price comparisons around the policy transition



Source: Eurostat HICP and national statistical institutes (Panel A); Statistics Denmark (Panel B).

Notes: Panel A compares Danish log beef-and-veal HICP with the SDiD-weighted other-EU series; the donor level is shifted to Denmark’s pre-period mean for display only. Panel B displays the log beef-minus-food CPI gap relative to June 2024 without pointwise inference bars. Both have July 2024 as the first post month and end in September 2025. The country estimator uses synthetic pre-period time weights; the plotted paths alone do not give its contrast. Appendix Figure B3 shows the separate grocery change-event diagnostic.

The grocery change-event regression in Table 2, column 1, gives 0.102 (product–store clustered standard error 0.013) after a conservative beef-name screen and publication-window eligibility check. Its 151 beef product–store units contribute 1,105 recorded event-months. Because the dependent variable averages price changes and inclusion depends on repricing, neither the point estimate nor its small clustered standard error validates a monthly national posted-price effect. Appendix Figure B3 documents its event path. The extra-EU bilateral-import diagnostic gives -0.080 (pair-bootstrap standard error 0.078). Its bilateral pairs are not independent Danish policy episodes, and it cannot bound total foreign substitution (Appendix Figure B4).

The production analysis yields a main slaughter-weight ATT of -0.066 (placebo standard error 0.081), or a point estimate of about a 6.4 percent decline. Its 95 percent interval, $[-0.225, 0.093]$, includes zero. With a longer, pre-treatment seasonally adjusted panel, the estimate is -0.006 .

Slaughter counts show similarly uncertain responses. Direct estimation of log mean carcass weight on common country-months gives -0.018 (placebo standard error 0.026), with an interval spanning zero. Subtracting the separate weight and head-count ATTs would not estimate this outcome because their synthetic weights can differ. Appendix Table B2 presents the slaughter estimates and June-omission check, and Table B3 displays raw-log 15-, 24-, 36-, and 54-month estimates for every outcome. Production evidence does not establish the quantity contraction implied by the model, and the sensitivity across windows is relevant to interpreting the price results.

The negative sign of the main slaughter estimate is nevertheless consistent with the model’s predicted contraction. Its economic magnitude depends in part on supply elasticity: a less elastic production technology limits quantity adjustment to a given change in expected net returns, while more elastic supply permits a larger response. In the model, the inverse supply elasticity η enters together with demand elasticity and intertemporal adjustment costs. A small quantity response alongside rising prices could therefore be consistent with biological or capacity constraints, but the imprecise slaughter estimates do not establish that explanation.

The observed price-quantity combination does not by itself identify supply elasticity. The announcement shifts expected production costs, so the resulting equilibrium moves along demand as well as reflecting supply adjustment. Moreover, consumer prices and slaughterhouse output measure different stages of an internationally traded market. Taking the ratio of the slaughter ATT to the retail-price ATT would consequently not recover a supply elasticity. The evidence leaves both relatively rigid short-run supply and more responsive production compatible with the data; distinguishing them would require farm-level output and producer-price variation that isolates movements along supply.

The two-date model also disciplines magnitude. Even assigning the full 59.6 kg CO₂e/kg life-cycle intensity to the future average output burden, the 2024-DKK rate of $125.93/\text{tCO}_2\text{e}$ gives a wedge of about DKK $7.51/\text{kg}$. For $\delta \leq 1$, Corollary 2 limits the stationary two-date market-price response to DKK $3.75/\text{kg}$. Table 3 maps both official retail coefficients through identical illustrative price levels. At the revised provisional event-price mean of DKK $147.77/\text{kg}$, the national mapping is DKK $9.83/\text{kg}$ and exceeds the stylized bound; the country mapping is DKK $2.33/\text{kg}$ and does not. The direction of the apparent magnitude tension thus depends on the counterfactual. Neither comparison tests the model: the level anchor is not a validated posted-price mean, retail and producer stages differ, lifecycle emissions are broader than the tax base, and the policy share of either contrast is unknown.

Environmental-price scenarios. Appendix C retains the conditional SCC arithmetic and its source inventory. It assumes rather than identifies the cattle-policy share of either price contrast, taxable intensity, persistence, and later pass-through. The attributed share and persistence enter that arithmetic only as a product; restricting either to $[0, 1]$ is a scenario choice, not an identified causal bound. The exercise has no role in selecting the empirical counterfactual or establishing a welfare effect.

Table 3: Conditional mapping of official price contrasts into DKK/kg

Level-price scenario	Anchor, DKK/kg	Domestic CPI DiD	Country beef HICP SDiD
Illustrative low level	100.00	6.65	1.58
Provisional grocery event mean	147.77	9.83	2.33
Illustrative high level	200.00	13.30	3.16

Source: Stata calculations from Table 2, columns 2 and 4, and illustrative price levels.

Notes: Each mapping is $\bar{p}[\exp(\hat{\beta}) - 1]$, with full attribution assumed only for arithmetic. The stationary two-date upper bound is DKK 3.75/kg at $\delta = 1$ under the deliberately broad lifecycle-emissions output wedge of DKK 7.51/kg. The first and third anchors are scenarios, not estimates; the middle anchor is a change-event mean with unobserved availability. Consumer-price indices, the model’s market price, and taxable farm emissions are distinct objects.

6 Discussion of the Results

The most direct descriptive comparison for common European beef-market movements puts Denmark beside other countries’ prices for the same product. It yields a small, imprecise post-June divergence, larger in 2025 than in late 2024. The within-Denmark beef-versus-food contrast is larger and also concentrated in 2025, but it does not control for a common European beef shock. This contrast between two defensible benchmarks is the paper’s central empirical result. It constrains a claim of a large uniquely Danish increase without proving a null effect. The grocery change-event regression cannot resolve that uncertainty.

The production results help assess this interpretation. A contraction in Danish slaughter output relative to other EU countries would support one observable implication of the planned-supply model. The main estimate has that sign, but its uncertainty is substantial and the longer-pre-period estimate is close to zero. The evidence therefore does not confirm the joint prediction of higher prices and lower output. It also does not exclude that prediction. Slaughterhouse activity measures only one stage of production, while herd composition, live-animal trade, emissions intensity, and procurement arrangements may adjust on different horizons.

The country design still depends on the synthetic donors’ untreated path and may include spillovers. Its weighted baseline gives most weight to May and June 2024, after earlier cattle-policy news. The July cutoff is therefore a useful reporting convention for a post-June contrast, not a clean boundary between a wholly untreated and treated episode. The extra-EU import analysis examines one possible offset to domestic supply changes, but its interval includes relevant increases and decreases and it excludes intra-EU trade. It cannot establish the absence of foreign substitution.

The nitrate-derogation change is a particularly important limitation. It was announced before the livestock-tax agreement and took effect near the beginning of the post-period. Both measures can change expected cattle-production costs. Neither national food comparisons nor cross-country cattle comparisons distinguish their effects with these aggregate data. The broader Green Tripartite package also includes land-use and other agricultural measures. Interpreting the estimates as the isolated causal effect of a future emissions tax would require additional variation in exposure to these measures.

Appendix C translates selected SCC values into hypothetical beef-price units. The calculation varies assumed attribution, persistence, taxable share, and later pass-through. Its output does not measure a welfare gap or a uniquely appropriate Danish tax rate.

The model shows why credibility can matter before collection begins and gives an upper bound under its two-date assumptions. The country-HICP mapping stays below the illustrative bound while the domestic CPI mapping exceeds it. Neither is a test of the mechanism because both cross market stages, the expected liability change is unknown, and the livestock economy has multiple adjustment dates and trade margins. A quantitative explanation would require producer-price transmission, farm-level policy exposure, and credible measures of taxable emissions and herd adjustment.

Several measurement issues remain relevant. The grocery history selectively records changes and lacks historical availability; its package normalization is partly heuristic. Official CPI avoids that event-selection problem but has one treated series and a limited donor set. Livestock products, including poultry and eggs, are excluded from the food donor pool, yet substitution and exposure to other parts of the agreement can still affect remaining foods. The country HICP check changes the control group while retaining the Danish statistical provider. Finally, surveillance detected the affected dairy herd in September 2025, before the officially dated October BVD outbreak interval (Dansk Veterinærtidsskrift, 2025; Danish Veterinary and Food Administration, 2026). Ending the sample in September reduces exposure to the later outbreak without ensuring that cattle markets were unaffected in the final month.

7 Conclusion

This paper quantifies beef-price contrasts during a Danish cattle-policy transition. Relative to the same product in other EU countries, Denmark’s post-June beef-and-veal HICP contrast is small and imprecise, with more of the fitted divergence in 2025 than in late 2024. The domestic beef-versus-food CPI contrast is larger and concentrated in 2025, but its significance depends on the HAC bandwidth. The difference between these benchmarks is economically material; neither identifies the isolated emissions-tax announcement. The grocery change-event history does not supply an independent monthly posted-price check, and the slaughter estimates do not establish a corresponding contraction.

The April nitrate announcement and later implementation overlap the episode, European beef-price pressure was present in 2025, and the pre-July synthetic baseline already includes policy news. The two-date model supplies sign alternatives and a conditional magnitude bound, but the different official price mappings do not support a single calibrated interpretation. The appendix accounting exercises are hypothetical. Distinguishing mechanisms requires farm-level variation in expected tax and nitrate exposure, herd and producer-price measures, and complete trade flows.

References

- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, 111(12):4088–4118.
- Barrage, L. and Nordhaus, W. D. (2024). Policies, projections, and the social cost of carbon: Results from the DICE-2023 model. *Proceedings of the National Academy of Sciences*, 121(13):e2312030121.
- Bilal, A. and Känzig, D. R. (2026). The macroeconomic impact of climate change: Global versus local temperature. *Quarterly Journal of Economics*, 141(2):889–944.
- Cai, Y., Lenton, T. M., and Lontzek, T. S. (2016). Risk of multiple interacting tipping points should encourage rapid CO₂ emission reduction. *Nature Climate Change*, 6:520–525.
- Cai, Y. and Lontzek, T. S. (2019). The social cost of carbon with economic and climate risks. *Journal of Political Economy*, 127(6):2684–2734.
- Carleton, T. et al. (2022). Valuing the global mortality consequences of climate change accounting for adaptation costs and benefits. *Quarterly Journal of Economics*, 137(4):2037–2105.
- Danish Government (2024). Regeringen og parterne i grøn trepart indgår historisk aftale om et grønt danmark. Government announcement.
- Danish Ministry of Economic Affairs (2024). Regeringen indgår politisk aftale om grøn trepart. Press release, 18 November.
- Danish Ministry of Environment (2024). Kvægundtagelsen ophører 31. juli 2024. Press release, 10 April.
- Danish Ministry of Taxation (2024a). Ekspertgruppe klar med modeller for en grøn skattereform på landbrug. Press release, 21 February.
- Danish Ministry of Taxation (2024b). Faktaark: Skatteinitiativer i aftale om et grønt danmark. Government fact sheet.
- Danish Veterinary and Food Administration (2026). Bovin virus diarré (BVD).
- Dansk Veterinærtidsskrift (2025). Kampen mod de usynlige fjender hos kvæg.
- Dietz, S., Rising, J., Stoerk, T., and Wagner, G. (2021). Economic impacts of tipping points in the climate system. *Proceedings of the National Academy of Sciences*, 118(34):e2103081118.
- Eurostat (2026a). Harmonised index of consumer prices: Monthly indices for beef and veal. European Commission. Historical monthly consumer-price indices.

- Eurostat (2026b). Slaughtering in slaughterhouses: Monthly data. Monthly bovine slaughter statistics.
- Fabra, N. and Reguant, M. (2014). Pass-through of emissions costs in electricity markets. *American Economic Review*, 104(9):2872–2899.
- Hänsel, M. C. et al. (2020). Climate economics support for the UN climate targets. *Nature Climate Change*, 10:781–789.
- Moore, F. C. et al. (2024). Synthesis of evidence yields high social cost of carbon due to structural model variation and uncertainties. *Proceedings of the National Academy of Sciences*, 121(52):e2410733121.
- Nordhaus, W. D. (2017). Revisiting the social cost of carbon. *Proceedings of the National Academy of Sciences*, 114(7):1518–1523.
- OECD (2025). *Measuring Carbon Footprints of Agri-food Products*. OECD Publishing, Paris.
- Pindyck, R. S. (2019). The social cost of carbon revisited. *Journal of Environmental Economics and Management*, 94:140–160.
- Poore, J. and Nemecek, T. (2018). Reducing food’s environmental impacts through producers and consumers. *Science*, 360(6392):987–992.
- Rennert, K. et al. (2022). Comprehensive evidence implies a higher social cost of CO₂. *Nature*, 610:687–692.
- Ricke, K., Drouet, L., Caldeira, K., and Tavoni, M. (2018). Country-level social cost of carbon. *Nature Climate Change*, 8:895–900.
- Rittenhouse, K. and Zaragoza-Watkins, M. (2018). Anticipation and environmental regulation. *Journal of Environmental Economics and Management*, 89:255–277.
- Rosen, S., Murphy, K. M., and Scheinkman, J. A. (1994). Cattle cycles. *Journal of Political Economy*, 102(3):468–492.
- Smulders, S., Tsur, Y., and Zemel, A. (2012). Announcing climate policy: Can a green paradox arise without scarcity? *Journal of Environmental Economics and Management*, 64(3):364–376.
- Statistics Denmark (2026). Slaughterings and production of cattle. Monthly cattle production statistics.
- Teagasc (2025). Production declines to boost beef farm incomes in 2025.
- van den Bremer, T. S. and van der Ploeg, F. (2021). The risk-adjusted carbon price. *American Economic Review*, 111(9):2782–2810.

A Theory Appendix

A.1 Proof of Proposition 1

The interior first-order conditions for (3) are

$$G'_0(Q_0) + \delta\kappa(Q_T - Q_0) = 0, \quad (\text{A1})$$

$$G'_T(Q_T) - \omega - \kappa(Q_T - Q_0) = 0. \quad (\text{A2})$$

Define $h_0 \equiv -G''_0(Q_0) > 0$, $h_T \equiv -G''_T(Q_T) > 0$, and

$$\Delta \equiv (h_0 + \delta\kappa)(h_T + \kappa) - \delta\kappa^2 = h_0h_T + h_0\kappa + \delta\kappa h_T > 0.$$

Implicit differentiation of (A1)–(A2) gives

$$\begin{bmatrix} h_0 + \delta\kappa & -\delta\kappa \\ -\kappa & h_T + \kappa \end{bmatrix} \begin{bmatrix} \partial Q_0 / \partial \omega \\ \partial Q_T / \partial \omega \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Cramer's rule therefore yields

$$\frac{\partial Q_0}{\partial \omega} = -\frac{\delta\kappa}{\Delta} < 0, \quad \frac{\partial Q_T}{\partial \omega} = -\frac{h_0 + \delta\kappa}{\Delta} < 0. \quad (\text{A3})$$

Since $P'_0(Q_0) < 0$, $\partial P_0 / \partial \omega = P'_0(Q_0) \partial Q_0 / \partial \omega > 0$. The first expression in (A3) is zero when $\kappa = 0$ or $\delta = 0$; when $\pi = 0$, $\omega = \pi\tau\chi = 0$ and there is no policy-induced change. This proves the proposition.

A.2 Derivation of Corollary 1

For $\varepsilon \neq 1$, the marginal utility of beef in (5) is $Aq^{-1/\varepsilon}$; the logarithmic case gives the same expression at $\varepsilon = 1$. Because the numeraire's marginal utility and price both equal one, the household's interior first-order condition is $p = Aq^{-1/\varepsilon}$, which proves (6). Its elasticity is $-(dQ/dP)(P/Q) = \varepsilon$.

With capital fixed, the production constraint (7) requires $X = (Q/(BK^{1-\alpha}))^{1/\alpha}$. Multiplication by the input price v gives (8). Differentiation yields

$$C'(Q) = \frac{v}{\alpha}(BK^{1-\alpha})^{-1/\alpha}Q^{1/\alpha-1} = mQ^\eta, \quad \eta = \frac{1-\alpha}{\alpha} > 0.$$

Equating inverse demand and marginal cost at a static competitive equilibrium gives (9). At that equilibrium,

$$h \equiv -G''(Q^*) = -P'(Q^*) + C''(Q^*) = \frac{P^*}{Q^*} \left(\frac{1}{\varepsilon} + \eta \right) = \frac{P^*}{Q^*} s. \quad (\text{A4})$$

Stationarity and $\omega = 0$ imply $Q_0 = Q_T = Q^*$, so $h_0 = h_T = h$ in (A3). Since $\rho = \kappa Q^* / P^*$, substituting (A4) into (A3), multiplying quantity derivatives by P^* / Q^* , and using $\partial \log P / \partial \log Q = -1/\varepsilon$ gives (10)–(12). These are derivatives evaluated at the nonlinear model's stationary equilibrium,

so no linear demand or marginal-cost schedule is assumed.

Let

$$R(\delta, \rho) \equiv \frac{\delta \rho}{\varepsilon s[s + \rho(1 + \delta)]}$$

denote the coefficient in (12). Direct differentiation gives

$$\frac{\partial R}{\partial \delta} = \frac{\rho(s + \rho)}{\varepsilon s[s + \rho(1 + \delta)]^2} > 0, \quad \frac{\partial R}{\partial \rho} = \frac{\delta}{\varepsilon[s + \rho(1 + \delta)]^2} > 0.$$

Thus proximity and adjustment curvature strengthen local announcement incidence. The factors π , τ , and χ enter through $\hat{\omega} = \pi\tau\chi/P^*$.

A.3 Proof of Corollary 2

If $\kappa = 0$, the announcement-date first-order condition is $G'(Q_0) = 0$, hence $Q_0 = Q^*$ and the bound holds with equality at zero on its left-hand side. For $\kappa > 0$, write $x = G'(Q_0)$ and $y = G'(Q_T)$ under stationary primitives. The first-order conditions of (3) imply $x = \delta\kappa(Q_0 - Q_T)$ and $y = \omega - \kappa(Q_0 - Q_T)$, so $x + \delta y = \delta\omega$. If $Q_T \geq Q_0$, then $x \leq 0$ and $y \geq \omega > 0$, while strict decrease of G' implies $y \leq x$, a contradiction. Thus $Q_T < Q_0$, $y > x > 0$, and $x \leq \delta\omega/(1 + \delta)$. Since $G'(Q^*) = 0$, $Q_0 < Q^*$. Weakly increasing marginal cost then gives

$$P(Q_0) - P(Q^*) = x + C'(Q_0) - C'(Q^*) \leq x \leq \frac{\delta}{1 + \delta}\omega.$$

The proof uses the output wedge and price at the same market stage. Retail margins and imported supply are outside its assumptions.

A.4 Multi-period anticipation

Consider dates $t = 0, \dots, N$ and implementation at $T \leq N$. Let $\mathbf{1}\{\cdot\}$ denote the indicator function. The multi-period potential is

$$\sum_{t=0}^N \beta^t [G_t(Q_t) - \omega \mathbf{1}\{t \geq T\} Q_t] - \sum_{t=1}^N \beta^t \frac{\kappa}{2} (Q_t - Q_{t-1})^2. \quad (\text{A5})$$

Let $\mathcal{H}$ be its Hessian in $Q = (Q_0, \dots, Q_N)'$ and define $\mathcal{M} \equiv -\mathcal{H}$. The restrictions $G_t'' < 0$, together with the nonnegative quadratic adjustment terms, make $\mathcal{M}$ positive definite. Its diagonal entries are positive, its only off-diagonal entries are non-positive adjacent adjustment terms, and for $\kappa > 0$ it is irreducible. Hence $\mathcal{M}$ is a nonsingular tridiagonal M-matrix and $\mathcal{M}^{-1}$ is elementwise strictly positive. Differentiating the first-order conditions with respect to ω gives

$$\mathcal{M} \frac{\partial Q}{\partial \omega} = -b, \quad b_t = \beta^t \mathbf{1}\{t \geq T\} \geq 0.$$

Therefore $\partial Q/\partial \omega = -\mathcal{M}^{-1}b < 0$ at every date. A credible future wedge thus propagates backward through the production plan.

Under the stationary microfoundations in (5)–(8), evaluate the local response at the zero-wedge equilibrium $Q_t = Q^*$. For interior pre-implementation dates $1 \leq t \leq T-1$, differentiating the first-order conditions gives the homogeneous recurrence

$$\kappa\psi_{t-1} - [h + \kappa(1 + \beta)]\psi_t + \beta\kappa\psi_{t+1} = 0,$$

where $\psi_t \equiv \partial Q_t/\partial \omega$ and $h = (P^*/Q^*)s$ is given by (A4). Writing the stable component as $\psi_t \propto r_-^{T-t}$ gives the polynomial $\kappa r^2 - \mathcal{B}r + \beta\kappa = 0$, where $\mathcal{B} = h + \kappa(1 + \beta)$. This polynomial has roots $0 < r_- < 1 < r_+$. Selecting the smaller root is exact for a semi-infinite preperiod under boundedness as $T - t \rightarrow \infty$ and gives an interior approximation for a long finite preperiod:

$$r_- = \frac{\mathcal{B} - \sqrt{\mathcal{B}^2 - 4\beta\kappa^2}}{2\kappa} \in (0, 1), \quad \mathcal{B} \equiv h + \kappa(1 + \beta). \quad (\text{A6})$$

Thus the absolute magnitude of the negative stable component, which is proportional to r_-^{T-t} , declines geometrically with distance from implementation. On an arbitrary finite horizon, the exact path combines both characteristic roots, with coefficients fixed by the initial and terminal boundary conditions; equation (A6) is not an exact global finite-horizon solution. The M-matrix sign result above remains exact. The geometric characterization is the multi-period counterpart to the role of $\delta = \beta^T$ in the two-date model.

A.5 Welfare and the Pigouvian benchmark

Assume tax revenue is rebated lump-sum, tax administration is costless, private and social discount factors coincide, adjustment costs are real resource costs, and emissions intensity χ is fixed so that output is the only abatement margin. With emissions $E_t = \chi Q_t$, social welfare in the two-date model is

$$W = G_0(Q_0) - D_0(\chi Q_0) + \delta \left[G_T(Q_T) - D_T(\chi Q_T) - \frac{\kappa}{2}(Q_T - Q_0)^2 \right]. \quad (\text{A7})$$

At a given date, equating the private output wedge $\chi\tau_t$ to marginal environmental damage per unit of output $\chi D'_t(E_t)$ gives $\tau_t^P = D'_t(E_t) = MD_t^e$, proving (15). The paper's SCC comparison additionally imposes the one-for-one CO2e valuation convention stated in the main text.

Along the private equilibrium defined by (A1)–(A2), differentiation of (A7) and substitution of the first-order conditions yield

$$\frac{dW}{d\omega} = -\chi D'_0(E_0) \frac{\partial Q_0}{\partial \omega} + \delta [\omega - \chi D'_T(E_T)] \frac{\partial Q_T}{\partial \omega}. \quad (\text{A8})$$

Both quantity derivatives are negative by Proposition 1. At $\omega = 0$, $D'_0(E_0) > 0$, $D'_T(E_T) > 0$, $\kappa > 0$, and finite T make both terms strictly positive. More generally, $D'_0(E_0) > 0$ and $\omega \leq \chi D'_T(E_T)$ are sufficient for a positive local welfare effect. This proves Proposition 2. It is a second-best statement

about a future instrument: it does not imply that omission of a current tax is first-best.

A.6 Fixed-stock counterexample

Let a fixed stock $\mathcal{S} > 0$ of slaughterable output satisfy $Q_0 + Q_T = \mathcal{S}$, with $0 < Q_0 < \mathcal{S}$. The stock and output share the same units, and the tax liability is assumed to apply to later slaughtered output. Substituting $Q_T = \mathcal{S} - Q_0$, the private objective is

$$G_0(Q_0) + \delta[G_T(\mathcal{S} - Q_0) - \omega(\mathcal{S} - Q_0)].$$

Its first-order condition is $G'_0(Q_0) - \delta G'_T(\mathcal{S} - Q_0) + \delta\omega = 0$. Strict concavity gives

$$\frac{\partial Q_0}{\partial \omega} = -\frac{\delta}{G''_0(Q_0) + \delta G''_T(\mathcal{S} - Q_0)} > 0, \quad \frac{\partial P_0}{\partial \omega} < 0.$$

A future tax therefore accelerates slaughter when fixed stock cannot adjust, opposite to planned contraction. If emissions accrue continuously before slaughter, this two-date result need not apply.

B Additional Results

Table B1: Preferred SCC benchmarks and pooling status

Study	Method	Reference year	2024 USD/tCO2	Low	High	Pool status
Cai et al. (2016)	Stochastic IAM	2010	166.9	—	—	Included
Nordhaus (2017)	DICE IAM	2030	74.2	—	—	Included
Ricke et al. (2018)	Spatial IAM	2020	599.9	254.6	1,158.0	Included
Cai–Lontzek (2019)	Stochastic IAM	2005	54.3	—	—	Included
Pindyck (2019)	Expert elicitation	Undated	107.0	—	—	Included
Hänsel et al. (2020)	Updated DICE	2020	133.7	22.5	688.2	Included
van den Bremer–van der Ploeg (2021)	Asset-pricing IAM	2015	52.7	—	—	Included
Dietz et al. (2021)	Meta-analytic IAM	2020	78.5	—	—	Included
Rennert et al. (2022)	Modular IAM	2020	224.2	53.3	500.6	Included
Barrage–Nordhaus (2024)	DICE-2023	2030	112.5	—	—	Included
Carleton et al. (2022)	Mortality damages	2020	21.0	−30.3	65.8	Partial; external
Moore et al. (2024)	Literature synthesis	2020	343.0	38.8	1,059.3	Double-count risk; external
Bilal–Känzig (2026)	Macro empirical	2024	> 1,200	—	—	Lower bound; external

Source: The studies listed in the first column.

Notes: Values are converted to 2024 US dollars per ton of CO₂. One preferred global total-SCC estimate enters per eligible paper. Values at dates other than 2030 are held constant as an explicit sensitivity assumption. Pindyck estimates an average SCC; excluding it gives a mean of USD 166.32/tCO₂. Barrage–Nordhaus is log-interpolated to 2030. Low and high retain their source-specific interpretation. The pooled mean gives equal weight to each included paper; partial estimates, a literature synthesis, and a lower bound are external comparisons.

B.1 Cattle slaughter and production

The model’s production prediction is examined using Eurostat’s monthly slaughterhouse statistics for bovine meat. The primary outcome is the log of carcass weight in thousand tonnes; the second outcome is the log of slaughtered animals in thousand heads. Denmark is the treated unit and the other 26 EU Member States are donors. No other Danish meat category enters the comparison. The statistics measure activity in slaughterhouses (Eurostat, 2026b).

The main window matches the official price analysis: April 2023–September 2025, with July 2024 as the first post-announcement month. All 27 countries have positive, non-missing observations in each selected month. Missing values would exclude a country from a balanced specification; they are never replaced by zero. Placebo inference uses 200 replications. The analysis concerns slaughterhouse output, which can differ from domestic cattle production when live animals cross borders or slaughter takes place outside registered facilities. Statistics Denmark’s cattle production series provides a separate national production measure; its live-animal component must be distinguished from the Eurostat slaughterhouse measure (Statistics Denmark, 2026).

Two sensitivity checks examine the dependence on temporal support. The first starts in January 2020, giving 54 pre-treatment months. For each country, it subtracts the pre-treatment mean log output for the corresponding calendar month and adds back that country’s overall pre-treatment mean. This removes estimated country-specific seasonality using only pre-announcement observations. Because it changes both temporal support and seasonal adjustment, it is a joint sensitivity check. The second retains the shorter window but drops June 2024, leaving 14 pre-treatment months. Neither specification is selected on the sign or significance of its estimate.

Table B2: Synthetic DiD estimates for Danish bovine slaughter

	Log slaughter weight			Log slaughter counts		
	(1)	(2)	(3)	(4)	(5)	(6)
ATT	−0.066 (0.081)	−0.006 (0.080)	−0.030 (0.090)	−0.035 (0.081)	−0.023 (0.079)	0.002 (0.092)
Observations	810	1,863	783	810	1,863	783
Countries	27	27	27	27	27	27
Pre-treatment months	15	54	14	15	54	14
Post-treatment months	15	15	15	15	15	15
Long pre-period	No	Yes	No	No	Yes	No
Seasonal adjustment	No	Yes	No	No	Yes	No
June 2024 omitted	No	No	Yes	No	No	Yes

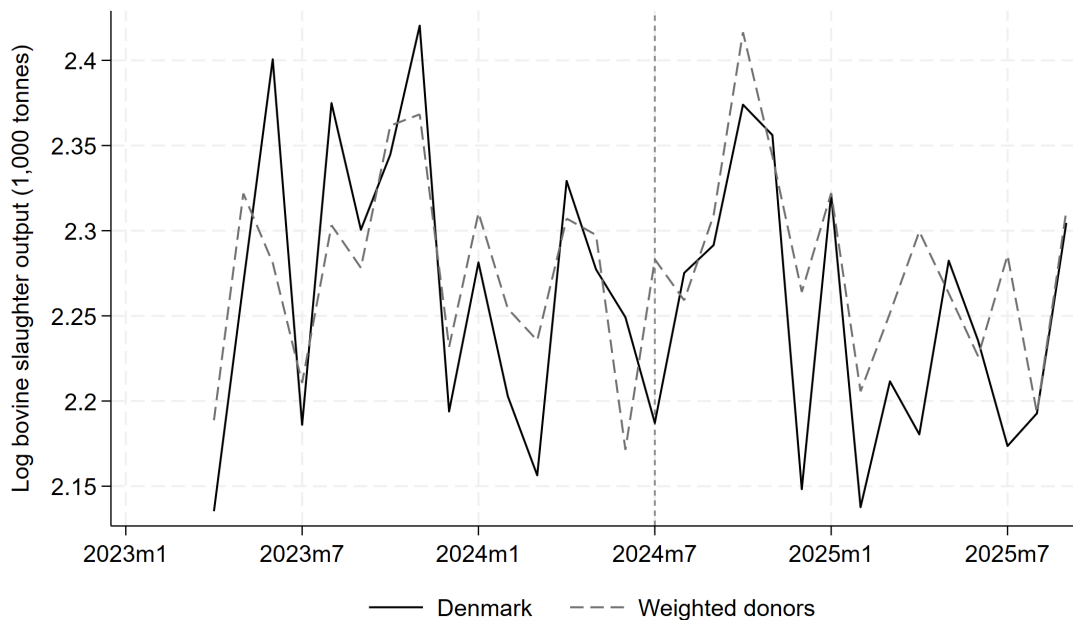
Source: Eurostat, monthly bovine slaughterhouse statistics.

Notes: Denmark is treated and the other 26 EU Member States are donors. Columns 1 and 4 use April 2023–September 2025; columns 2 and 5 begin in January 2020 and remove country-specific seasonality estimated before treatment; columns 3 and 6 omit June 2024. Parentheses contain placebo standard errors based on 200 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. No estimate reaches these significance thresholds.

The main weight estimate is negative, but the interval allows a substantial decline as well as an increase. Its pre-treatment centered gap root-mean-square error is 0.056 log units. The longer-pre-

period specification has an error of 0.031 and an ATT close to zero. The ratio of slaughter weight to slaughtered heads directly measures registered mean carcass weight on common source cells. Estimating its log with the same 27-country, 30-month design gives an ATT of -0.018 (placebo standard error 0.026; 95 percent interval $[-0.069, 0.032]$). This estimate cannot be obtained by subtracting columns 1 and 4 because the two SDiD fits need not use identical weights. The uncertain responses limit evidence for a contraction or a shift in carcass weight. In addition, the end of the Danish cattle nitrate derogation in July 2024 affects the interpretation of every specification. Country controls absorb common market movements but cannot separate that domestic regulatory change from the tax-announcement episode.

Figure B1: Danish bovine slaughter weight and synthetic-DiD comparison



Source: Eurostat, monthly bovine slaughter weight in thousand tonnes.

Notes: The main sample covers April 2023–September 2025. The dashed vertical line marks July 2024. For display, the donor series is shifted by a constant to match Denmark’s pre-treatment mean; this does not change the SDiD estimate. The ATT additionally uses time weights; the plotted level gap alone is not the estimated treatment effect. Production estimates and uncertainty appear in Table B2.

B.2 Pre-period support sensitivity

Table B3 varies the number of pre-treatment months while holding the July 2024–September 2025 post-period and each outcome’s donor set fixed. The slaughter outcomes here remain in raw logs; the seasonally adjusted 54-month specification in Table B2 is a separate joint sensitivity. Fit is measured on the common April 2023–June 2024 pre-period after removing the mean treated–synthetic gap, as permitted by synthetic DiD’s intercept. Longer support slightly improves this in-sample fit for country HICP at 36 and 54 months, while worsening it for the other outcomes.

No outcome’s preferred window is chosen by another outcome’s fit. These point estimates use no-inference diagnostic runs, so the table does not supply new standard errors. In-sample fit cannot establish the untreated post-period path; a prespecified held-out pre-period and donor-weight audit would provide stronger validation.

Table B3: Synthetic-DiD window sensitivity by outcome

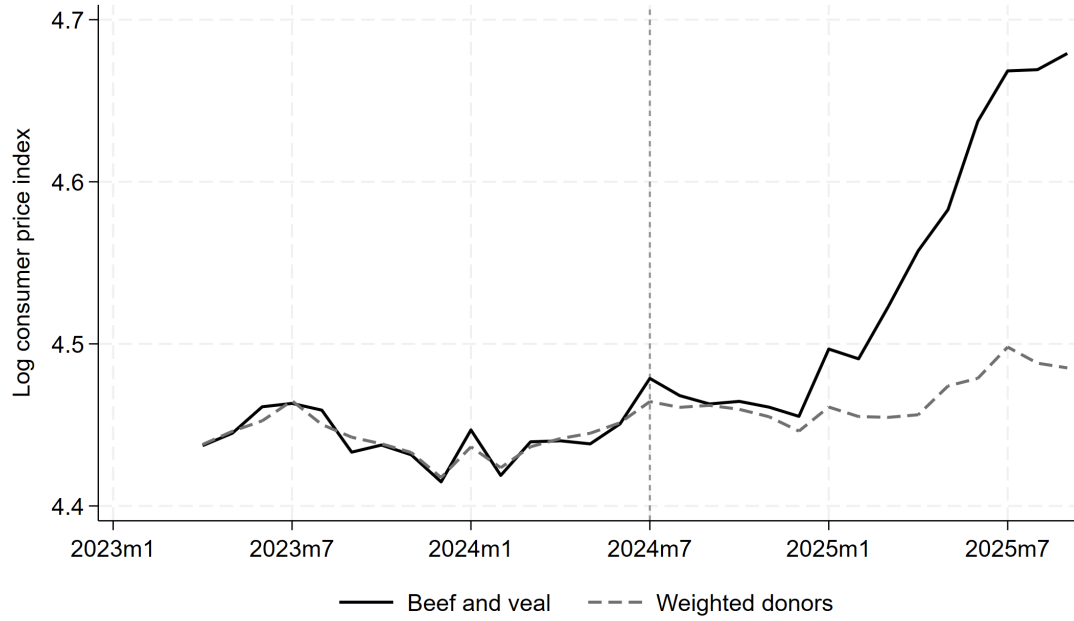
Pre-treatment months	15	24	36	54
<i>Panel A. Point ATT</i>				
Danish beef versus food CPI	0.076	0.043	0.062	0.060
Denmark versus EU beef HICP	0.016	0.028	0.023	0.022
Slaughter weight	−0.066	−0.078	−0.021	−0.014
Slaughter heads	−0.035	−0.053	−0.009	−0.009
Extra-EU bilateral imports	−0.080	−0.077	−0.079	−0.084
<i>Panel B. Centered pre-gap RMSE on common 15 months</i>				
Danish beef versus food CPI	0.005	0.010	0.012	0.011
Denmark versus EU beef HICP	0.010	0.010	0.010	0.010
Slaughter weight	0.056	0.060	0.062	0.063
Slaughter heads	0.056	0.061	0.063	0.064
Extra-EU bilateral imports	0.065	0.080	0.086	0.096

Source: Statistics Denmark, Eurostat, and European Commission data.

Notes: All specifications retain their respective published donor sets. The 15-, 24-, 36-, and 54-month pre-periods all end in June 2024. ATT values are Stata no-inference diagnostics; Panel B evaluates every fit on the same April 2023–June 2024 months and permits a constant gap. Slaughter entries do not remove seasonality. The extra-EU outcome is $\log(1+\text{tonnes})$, not total imports.

B.3 Additional price and trade comparisons

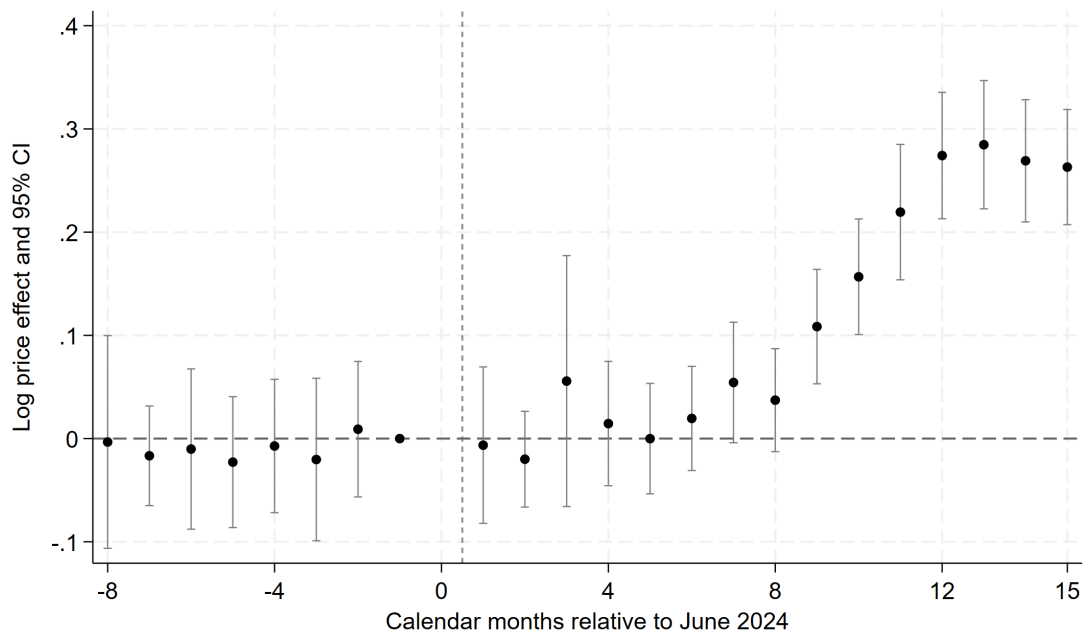
Figure B2: Official-data synthetic DiD: beef and weighted donors



Source: Statistics Denmark.

Notes: Series are log CPI for beef and veal and the SDiD-weighted donor series over the symmetric 15-month pre/post window. The dashed vertical line marks July 2024, the first post-announcement month. For display, the donor series is shifted by a constant to match the treated series' pre-treatment mean; this does not change the SDiD estimate.

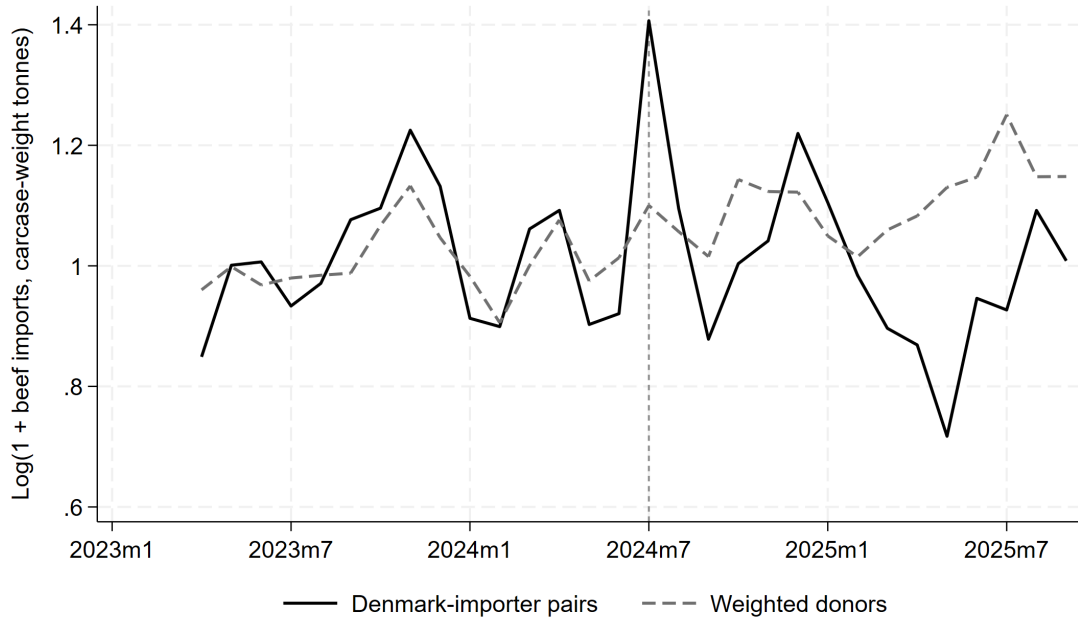
Figure B3: Recorded grocery change-event price diagnostic



Source: dagligepriiser.dk.

Notes: The plot uses 151 beef product-store units with within-window observations on both sides of June 2024. It averages recorded price-change events in each observed month; availability and unchanged-price spells are unknown. June is excluded, with May (−1) the reference and an empty slot at zero; July is +1. Bars are product-store clustered intervals conditional on event selection, not uncertainty for a national monthly posted-price response.

Figure B4: Synthetic DiD: beef imports into Denmark



Source: European Commission, Beef Trade Data Explorer; Eurostat COMEXT.

Notes: Series are mean log-one-plus monthly extra-EU beef imports, measured in carcass-weight tonnes, for Denmark-partner pairs and their SDiD-weighted donor series over the symmetric April 2023–September 2025 window. The dashed vertical line marks July 2024. The ATT is -0.080 (unit-cluster bootstrap standard error 0.078; $p = 0.306$; 200 replications). For display, the donor series is shifted by a constant to match the treated series' pre-treatment mean; this does not change the SDiD estimate.

C Conditional Environmental-Price Accounting

The price estimates do not measure how much climate damage consumers internalize. I instead ask how a hypothesized policy-attributable price increment compares, in DKK/kg, with a selected environmental-damage benchmark. The exercise requires separate assumptions about causal attribution, persistence until 2030, additional price adjustment at implementation, and the taxable emissions base. I construct a literature benchmark, express it in beef-price units, and keep the announcement and implementation components distinct. This is conditional arithmetic, not a welfare estimate or an estimate of optimal retail pricing.

The benchmark gives equal weight to one selected estimate from each of ten studies (Cai et al., 2016; Nordhaus, 2017; Ricke et al., 2018; Cai and Lontzek, 2019; Pindyck, 2019; Hänsel et al., 2020; van den Bremer and van der Ploeg, 2021; Dietz et al., 2021; Rennert et al., 2022; Barrage and Nordhaus, 2024). This is a documented selection of the literature rather than an exhaustive systematic review. The studies differ in discounting, damage functions, risk, and the object valued. In particular, Pindyck (2019) estimates an average cost of avoiding catastrophic outcomes, which differs from the marginal SCC in the other studies. Excluding that estimate raises the pooled benchmark to USD 166.32/tCO₂. Appendix Table B1 records the chosen estimates and the external comparisons.

Cai et al. (2016) reports both its initial SCC and monetary values for 2010. The inventory treats Pindyck’s monetary base as 2017, an explicit valuation-date assumption; its exclusion sensitivity addresses this additional comparability issue. I inflate dollar values to 2024 US dollars and multiply values per ton of carbon by 12/44 to obtain values per ton of CO₂. The Cai–Lontzek input is the combined economic-and-climate-risk value of USD 124/tC for 2005 in their Table 9, with intertemporal elasticity 1.5 and risk aversion 10. The Barrage–Nordhaus 2030 value is log-interpolated between published 2025 and 2050 points. Only this value and Nordhaus’s reported 2030 estimate refer directly to 2030; their two-study mean is USD 93.36/tCO₂. The remaining values are held constant for the calibration. This is an imposed sensitivity assumption even when a study supplies a rising or stochastic future path. The ten-study mean should therefore be interpreted as a constant-value benchmark used in a 2030 exercise, rather than a pooled forecast for that year.

The equal-paper-weight point mean across ten studies is USD 160.39/tCO₂. A sensitivity envelope is constructed in two explicit layers. Index papers by j . Let θ_j be paper j ’s preferred harmonized SCC estimate and $[L_j, U_j]$ its reported positive central range. Let $1 - 2\alpha_j$ denote its reported coverage where available, or an assumed coverage for a parameter span. Let Φ denote the standard normal cumulative distribution function and define

$$z_j = \Phi^{-1}(1 - \alpha_j), \quad \sigma_j = \frac{\log U_j - \log L_j}{2z_j}.$$

Here z_j is the corresponding standard-normal quantile and σ_j is the calibrated log-scale standard

deviation. For a selected paper with a usable range, the within-paper draw is

$$X_j = \exp(\log \theta_j - \frac{1}{2}\sigma_j^2 + \sigma_j Z_j), \quad Z_j \sim N(0, 1),$$

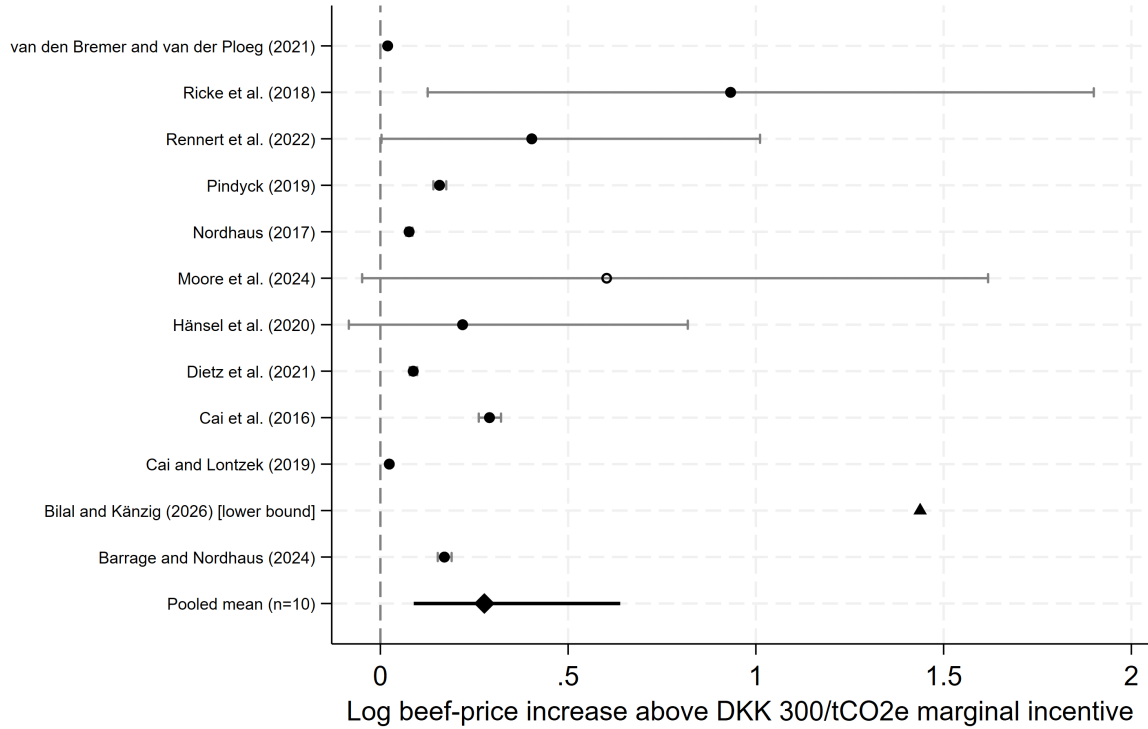
where Z_j is an independent standard-normal variate, X_j is the simulated SCC value, and $E[\cdot]$ denotes expectation. The $-\sigma_j^2/2$ adjustment ensures that $E[X_j] = \theta_j$, while the range's log width determines dispersion. Matching the arithmetic mean in this way generally means that L_j and U_j are not the simulated distribution's corresponding quantiles; the procedure matches their log width, not both endpoints. For papers without a positive range, $X_j = \theta_j$. The 66 percent model range in Ricke et al. (2018) has $\alpha_j = 0.17$ and therefore $z_j = \Phi^{-1}(0.83)$; the 5th–95th quantiles in Rennert et al. (2022) have $\alpha_j = 0.05$ and $z_j = \Phi^{-1}(0.95)$; and the ethical-parameter range in Hänsel et al. (2020) is assigned 95 percent coverage as an explicit sensitivity assumption, so $\alpha_j = 0.025$ and $z_j = \Phi^{-1}(0.975)$.

In the second layer, each of 10,000 bootstrap replications samples ten papers with replacement from the ten-paper pool, makes an independent within-paper draw for every sampled paper, and records the arithmetic mean of those ten draws. Resampling papers describes sensitivity to the selected pool; the lognormal step adds dispersion calibrated to source ranges. Shared models and assumptions across studies mean that paper draws should not be interpreted as independent scientific replications. The 2.5th and 97.5th percentiles of the 10,000 replicated means are USD 79.50 and USD 366.63/tCO₂. Because the inputs mix model, parameter, and distributional uncertainty rather than comparable sampling errors, this interval is a hierarchical sensitivity envelope, not a frequentist confidence interval for a common structural parameter.

Figure C1 shows each preferred estimate after mapping it into a hypothetical log beef-price equivalent net of the DKK 300/tCO₂e marginal abatement incentive. This marginal rate is informative about emissions-reduction incentives; it is not the average burden on beef output. Filled points enter the pool. The hollow point is the external literature synthesis, whose inclusion could double count underlying studies (Moore et al., 2024). The partial mortality SCC is reported separately in Appendix Table B1 (Carleton et al., 2022). The triangle is the lower bound reported by Bilal and Känzig (2026) and also remains external. The wide intervals reveal why retaining study-level variation materially changes the accounting comparison.

Let $\bar{p} = 147.77$ be the provisional mean pre-announcement beef change-event price in DKK/kg, $r^{FX} = 6.895$ DKK/USD the exchange rate, and $\tilde{\chi} = 59.6$ kg CO₂e/kg beef the lifecycle carbon-intensity benchmark. The benchmark follows the OECD's presentation of the beef-herd estimate from Poore and Nemecek (2018) (OECD, 2025). It includes lifecycle emissions beyond the farm-level emissions base to which the Danish levy applies. Equation (C1) is therefore a deliberately broad accounting sensitivity and can overstate the price-equivalent wedge if $\tilde{\chi}$ is interpreted as taxable emissions intensity. Neither the OECD presentation nor the accompanying source tables provide a sampling interval for this specific intensity benchmark. Their producer-level percentiles describe heterogeneity across production systems, rather than uncertainty in the mean used here. I therefore hold $\tilde{\chi}$ fixed; the scenarios for f below address the assumed taxable share, without

Figure C1: Selected SCC estimates and hypothetical beef-price equivalents



Source: The studies listed in Appendix Table B1; OECD (2025).

Notes: Each point maps a preferred SCC estimate into the log beef-price increase needed above the DKK 300/tCO₂e marginal incentive, using a 59.6 kg CO₂e/kg lifecycle benchmark, 6.895 DKK/USD, and the scraped pre-announcement beef price. The quoted 2022-DKK statutory rate is converted to 2024 DKK using the annual net-price index. The lifecycle benchmark is broader than the farm-level emissions base subject to the Danish levy, so the mapping is an accounting sensitivity rather than a tax-base calculation. Bars show simulated 2.5th–97.5th percentile sensitivity intervals combining available SCC ranges with resampling of the grocery price benchmark. SCC values without a usable range are held fixed; their bars capture price resampling alone. The lower-bound triangle is conditional on the central price benchmark and has no simulated interval. Original source ranges remain in Appendix Table B1. Filled circles enter the equal-paper-weight pool; hollow circles and the lower-bound triangle do not. The pooled diamond maps the mean SCC; its line combines hierarchical SCC and price-benchmark draws.

assigning it a probability distribution.

The grocery level anchor is itself estimated from selected change-event months. I resample complete product–store histories and common circular blocks of two consecutive pre-announcement months, recomputing the observation-weighted mean in each replication. This preserves repeated observations within products and allows for shared monthly movements. The 405 observations cover 151 product–store pairs but only eight months. The resulting 2.5th–97.5th percentile range is DKK 131.46–166.28/kg. This range is only resampling sensitivity: it cannot correct the missing availability history, selective coverage, or package mismeasurement. One- and four-month blocks give similar ranges. I pair these draws independently with the SCC draws when constructing the mapped intervals in Figures C1 and C2. At the individual-study level, the same lognormal range calibration is used where available, including the external synthesis; other SCC point estimates are held fixed. Thus these bars need not reproduce the original source ranges, which remain reported in the appendix.

For SCC value c^{SCC} in USD/tCO₂ and Danish rate τ in DKK/tCO₂, define the additional log price required as

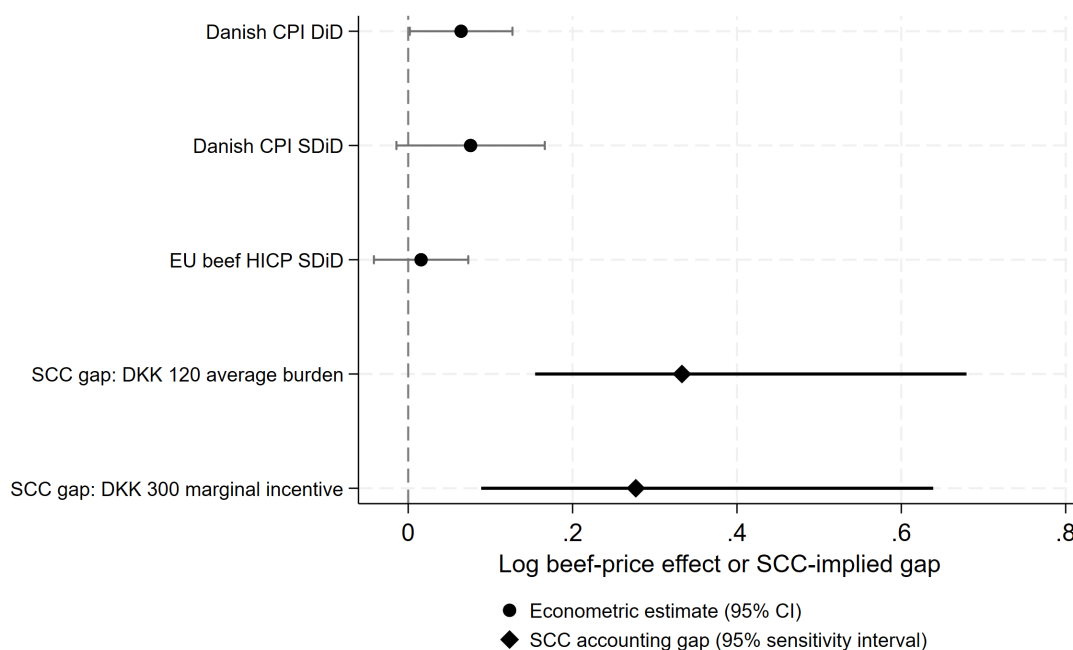
$$g(c^{SCC}; \tau) = \log \left[\frac{\bar{p} + (c^{SCC} r^{FX} - \tau) \tilde{\chi} / 1000}{\bar{p}} \right]. \quad (C1)$$

Here c^{SCC} denotes the SCC input. The quoted statutory rates are DKK 120 and DKK 300 in 2022 prices. Annual Danish net-price indices imply a 2024/2022 factor of 1.04938; hence τ is evaluated at 125.93 and 314.81 in 2024 DKK. The former is the effective average output burden created by the 60 percent per-animal deduction; the latter is the 2030 marginal abatement incentive (Danish Government, 2024; Danish Ministry of Taxation, 2024b). At the central calibration, $c^{SCC} = \bar{c}^{SCC} \equiv J^{-1} \sum_{j=1}^J E[X_j]$, where $J = 10$ is the number of eligible papers. Because the within-paper draws are centered so that $E[X_j] = \theta_j$, $\bar{c}^{SCC}$ is the equal-paper-weight mean. In each bootstrap replication, c^{SCC} is replaced by that replication’s arithmetic mean, and the price benchmark is replaced by its paired resampled mean.

At the pooled mean, the hypothetical log-price equivalent is 0.333 under the average-burden benchmark, with a sensitivity range of 0.154–0.679. Under the marginal-incentive benchmark, it is 0.277, with a range of 0.089–0.639. Figure C2 places these accounting magnitudes alongside the official econometric estimates. The ATT intervals retain their econometric interpretation, while the accounting ranges combine SCC and provisional grocery-level sensitivity. Their overlap is not a formal test of the difference between the ATT and the calibrated equivalent. Across leave-one-study-out calculations, the SCC mean ranges from USD 111.56 to USD 172.36/tCO₂, showing the influence of the highest estimate on the central benchmark.

To connect the calculation to the question posed at the start of this section, let $q \in [0, 1]$ be the assumed share of the observed price contrast attributable to the cattle-policy episode, $a \in [0, 1]$ the fraction of that attributed increment persisting until implementation, $f \in [0, 1]$ the fraction of lifecycle emissions used as a taxable-intensity approximation, and $\lambda \in [0, 1]$ additional implementation pass-through relative to the reference-animal average burden. The remaining hypothetical

Figure C2: Observed price contrasts and hypothetical SCC price equivalents



Source: Statistics Denmark; dagligepriser.dk; the studies listed in Appendix Table B1.

Notes: Points show the three official-price contrasts in Table 2, columns 2–4 and the SCC-implied log gaps under the DKK 120/tCO₂e effective average output burden and DKK 300/tCO₂e marginal abatement incentive. Quoted statutory rates are in 2022 DKK and are converted to 2024 DKK before calculation. Horizontal bars are 95 percent econometric intervals or the mapped 2.5th–97.5th percentile sensitivity interval combining hierarchical SCC draws and independent resampled grocery-price means. The ATT intervals do not depend on the level-price benchmark and retain their original standard errors. The 59.6 kg CO₂e/kg lifecycle benchmark is broader than the levy tax base. These are accounting comparisons, not estimates of the taxable emissions base, tax incidence, or welfare.

level difference in DKK/kg is

$$R(q, a, f, \lambda) = \frac{\bar{c}^{SCC} r^{FX} \tilde{\chi}}{1000} - qa\bar{p}(e^{\hat{\beta}} - 1) - \lambda \tau_{2024}^{avg} \frac{f \tilde{\chi}}{1000}, \quad (C2)$$

where $\hat{\beta} = 0.064$ is the domestic CPI DiD contrast in column 2 of Table 2 and $\tau_{2024}^{avg} = 125.93$. The parameter q is not estimated and refers to the cattle-policy episode, not to emissions-tax news alone. Restricting it to $[0, 1]$ is a scenario convention, not a causal bound: an omitted shock and a policy effect can have opposite signs. Likewise q and a enter only through their product, so the arithmetic does not identify attribution and persistence separately. The last term represents an *additional* price increment after the announcement period. If a pass-through estimate instead measures total incidence including earlier adjustment, that total must replace the two price terms together. This distinction prevents counting the same adjustment twice. The marginal abatement rate is excluded from this output-price decomposition because the fixed deduction separates marginal emissions incentives from the average output burden.

The CPI estimate is a proportional contrast rather than a DKK/kg change. Applying it to $\bar{p}$ assumes that the same proportional movement applies to the selected grocery change-event level. At $q = 1$, this maps the domestic CPI contrast to DKK 9.83/kg, while the pooled lifecycle-damage benchmark is DKK 65.91/kg. Under full attribution and persistence with no additional implementation increase, the hypothetical difference is DKK 56.09/kg. With full additional pass-through and the broad assumption $f = 1$, it falls to DKK 48.58/kg. At $q = 0$ with the same full additional pass-through and taxable share, it is DKK 58.41/kg. Thus attribution is a separate substantive assumption, not absorbed into persistence.

Figure C3 sets $q = 1$ to show a full-attribution scenario, varies announcement persistence a and additional implementation pass-through λ over $[0, 1]$ in four panels, and holds taxable share $f \in \{0.25, 0.50, 0.75, 1\}$ by panel. Each surface is a plane: persistence subtracts the same assumed attributed increment at every pass-through value, while the slope with respect to pass-through becomes steeper as the taxable share rises. At $q = 0$, persistence has no effect on the calculation. A narrow tax base limits the hypothetical additional price signal even when pass-through is complete. The panels describe conditional scenarios, not parameter forecasts.

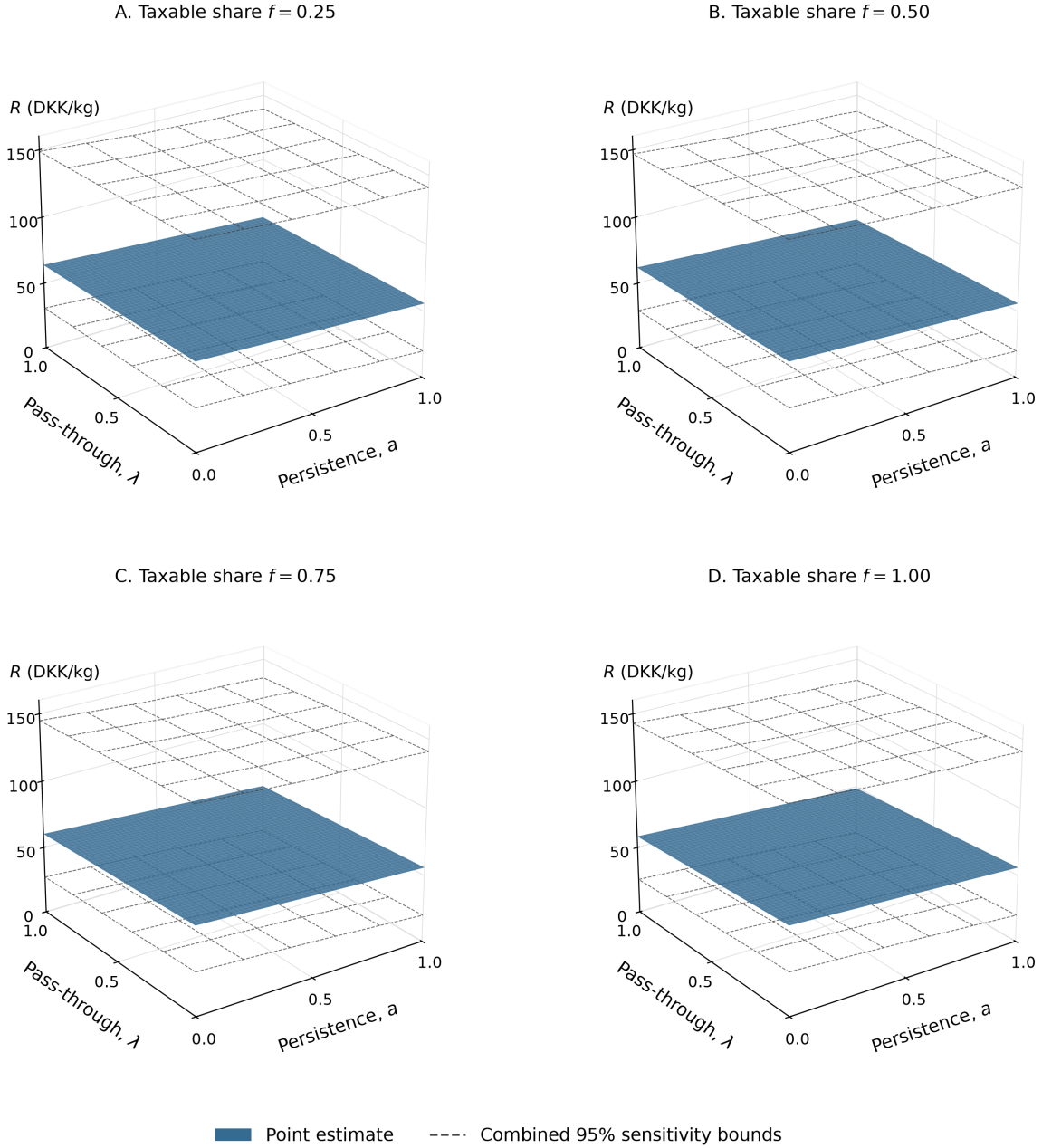
The dashed surfaces combine uncertainty in the preferred price coefficient, the hierarchical SCC variation, and the resampled grocery price benchmark. For each of the 10,000 SCC draws, I independently draw a price coefficient centered on the preferred estimate, using its HAC standard error and a Student t distribution with 28 degrees of freedom, matching the reference distribution used for Table 2. I also substitute a resampled price mean into equation (C2), then report the 2.5th and 97.5th percentiles of the resulting gap at each parameter combination. Independence across these three draws is an explicit simulation assumption. In particular, covariance between the grocery benchmark and the official-price estimate is not estimated, although both may respond to common market shocks. Emissions intensity, the exchange rate, and the policy rate remain fixed. The resulting bounds are pointwise combined sensitivity intervals, conditional on these

assumptions, rather than frequentist confidence bands. At $a = 0$, both coefficient and price-benchmark uncertainty drop out, while SCC variation remains.

Even at $q = a = f = \lambda = 1$, the combined 2.5th–97.5th percentile sensitivity range for the hypothetical difference is DKK 12.53–133.79/kg. Its width reflects variation in selected damage estimates, the price coefficient, and the provisional level anchor; it is not a sampling interval for a common SCC. Alternative month-block lengths give similar bounds. The smallest central difference in the stated scenario cube is DKK 48.58/kg, so none of these combinations closes the arithmetic difference at the selected damage benchmark. Under full attribution, persistence and additional pass-through, the two assumed price increments sum to about 26 percent of that benchmark. With no additional implementation response, the mapped CPI increment is about 15 percent. At full attribution and pass-through, equality would require an SCC of about USD 42.17/tCO₂; a smaller taxable share or weaker attribution lowers that threshold. These calculations do not imply that observed prices internalize environmental damages.

The exercise illustrates how strongly the arithmetic depends on attribution, the selected SCC studies, emissions intensity, the retail level anchor, and a possible later price response. Lifecycle CO_{2e} combines gases with different timing and damage profiles; the pre-announcement grocery mean spans several months rather than a single 2024 valuation date; and Danish production is linked to international markets. The calculation uses a common real-price approximation and a one-for-one CO_{2e} valuation convention. It cannot establish that any observed price movement represents internalized climate damages or identify the optimal Danish tax.

Figure C3: Hypothetical environmental-price difference at implementation, full attribution



Source: Statistics Denmark; dagligepriser.dk; the SCC studies in Appendix Table B1.

Notes: The vertical axis gives R in DKK/kg from equation (C2), setting attribution share $q = 1$ and using the domestic CPI DiD in Table 2, column 2. Panels vary persistence a and additional pass-through λ , holding taxable share f at the indicated value. Solid surfaces show point estimates. Dashed meshes show pointwise 2.5th–97.5th percentile bounds combining hierarchical SCC draws, resampled grocery-price means, and coefficient draws using the HAC standard error and Student $t(28)$ distribution. Independence across these sources is assumed. These are conditional sensitivity intervals, not sampling confidence bands for a common SCC or simultaneous confidence bands. Intensity, exchange rate, and tax rate stay fixed; the sources provide no sampling interval for the specific intensity benchmark. All panels share the same vertical scale.